Preservation of Q-energy under quasiconformal homeomorphisms of Ahlfors Q-regular limit spaces of admissible inverse systems

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A dissertation submitted in partial fulfillment

of the requirements for the degree of

Doctor of Philosophy

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New York University

August 2026

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Acknowledgements

The author thanks his advisor B. Kleiner for his support throughout the course of this dissertation and for many helpful discussions. The author is grateful for his encouragement and assistance in bringing this work to completion.

Abstract

We prove that bi-Lipschitz homeomorphisms between measurable subsets of X, an Ahlfors Q-regular limit space of an admissible inverse system, preserves Q-energy of Lipschitz functions by pullback, from which a standard argument implies that the same result holds for quasiconformal homeomorphisms between open subsets of X. Corollaries include preservation of Q-energy for bounded Newtonian functions and preservation of Q-harmonic functions. For a certain class of X constructed in a self-similar manner, we use the preservation of Q-harmonic functions to prove rigidity results for quasiconformal homeomorphisms between open subsets; in particular, such mappings are always local homotheties and can actually be constructed as inverse limits of homomorphisms between subgraphs of the graph approximations of X. We discuss applications of these ideas to the boundaries of certain right-angled Fuchsian buildings.

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1 Introduction

Doubling metric measure spaces admitting Poincaré inequalities have a function theory analogous to the function theory of Euclidean spaces. One may define a differentiable structure on these spaces for which the conclusion of Rademacher's theorem, that Lipschitz functions are differentiable a.e., holds ([7]). One may also develop a theory of Sobolev functions on such spaces ([7], [21], [22]).

Of particular interest are those Ahlfors Q-regular spaces admitting $(1, Q)$-Poincaré inequalities, also known as Loewner spaces. These spaces have a theory of quasiconformal maps like that of Euclidean spaces ([11]). It was originally not known for which Q such spaces existed. In [3], it is shown that such spaces exist with Q ranging over a dense set in $(1, \infty)$ as Gromov boundaries of 2-dimensional hyperbolic buildings. Such spaces for any given $Q \in (1, \infty)$ are constructed in [16], called "Laakso spaces." It was known that Laakso spaces could be obtained as limits of sequences of graphs. In [8], axioms for inverse systems of graphs are introduced whose limit spaces generalize Laakso spaces and provide a plethora of examples of PI spaces. Inverse systems that satisfy these axioms are termed "admissible inverse systems."

Our objective is to investigate certain properties of Lipschitz functions on Ahlfors Q-regular limit spaces of admissible inverse systems pertaining to Q-energy, in particular how the Q-energy behaves under pullback by quasiconformal maps. We apply this investigation to a specific class of inverse systems constructed in a self-similar manner and prove rigidity results for them pertaining to quasiconformal maps. We also discuss applications of some of these ideas to the boundaries of certain 2-dimensional hyperbolic buildings. Precise statements are given in Section 1.2.

1.1 History and background of rigidity

One form of the Mostow rigidity theorem [19] states that, given two complete hyperbolic manifolds of finite volume and dimension ≥ 3 (not a priori equal), then any isomorphism between their

fundamental groups is induced by a unique isometry. Mostow's proof of this theorem is based on quasiconformal analysis on the boundary at infinity of hyperbolic spaces. There are numerous generalizations or analogues of this theorem to other nonpositively curved spaces, in particular to symmetric spaces of noncompact type. Developments in analysis on metric spaces and quasiconformal geometry have been used to prove Mostow-type rigidity theorems in more general contexts.

Recall that a K-quasi-isometry $f : X \to Y$ between metric spaces is a mapping such that

$$K^{-1}d(x, y) - K \leq d(f(x), f(y)) \leq Kd(x, y) + K$$

and such that the K-neighborhood of the image $f(X)$ is equal to Y. Two quasi-isometries are equivalent if they are within bounded distance of one another. For finitely generated groups, we use their word metric, which defines a class of metrics equivalent up to quasi-isometry based on the choice of generating set. If a finitely generated group acts in a proper, discontinuous, and cocompact way on a proper metric space, e.g., the action of the fundamental group of a closed Riemannian manifold on its universal cover, then the group is quasi-isometric to the space on which it acts.

Given an increasing homeomorphism $\eta : [0, \infty) \to [0, \infty)$, we say that a homeomorphism $f : X \to Y$ is η-quasisymmetric if

$$\frac{d(f(x), f(y))}{d(f(x), f(z))} \leq \eta\left(\frac{d(x, y)}{d(x, z)}\right)$$

for all triples of distinct points x, y, z in X. Gromov hyperbolic metric spaces, which include the rank 1 symmetric spaces of noncompact type, have a class of metrics on their boundary at infinity equivalent up to quasisymmetry. The relationship between quasi-isometries and quasisymmetries in the setting of Gromov hyperbolic metric spaces is that equivalent quasi-isometries bijectively

correspond to quasisymmetric homeomorphisms between boundaries at infinity. This can be seen as a coarse version of the fact that, in the unit ball model of hyperbolic space, conformal mappings of the boundary sphere bijectively correspond to isometries of the interior.

A homeomorphism $f : X \to Y$ between metric spaces is said to be H-quasiconformal, for a constant $H \geq 1$, if

$$\limsup_{r \to 0} \frac{\sup \{d(f(x), f(y)) : d(x, y) \leq r\}}{\inf \{d(f(x), f(y)) : d(x, y) \geq r\}} \leq H.$$

Quasisymmetric implies quasiconformal and the converse is true locally in the context of Loewner spaces.

A few problems which are central in geometric group theory are as follows ([14], Question 2.3):

- What are the groups quasi-isometric to a given group?
- What is the quasi-isometry group of a given group?
- Are there conditions to determine if a group is quasi-isometric to a given model group?

Developments in quasiconformal analysis on the boundary can be seen as potential approaches to these problems in the setting of Gromov hyperbolic groups.

Pansu proved that a quasi-isometry of a quaternionic hyperbolic space or the Cayley plane is within bounded distance of a unique isometry [20]. The quasiconformal structure of the boundaries of these spaces are modeled on certain Carnot groups with their Carnot–Carathéodory metrics along with points at infinity, so Pansu's result can be restated as quasiconformal homeomorphisms of these boundaries extending a unique isometry. There is no analogue in the real hyperbolic space case, as quasiconformal mappings of the boundary sphere exist in infinite-dimensional families.

Capogna and Cowling proved that a 1-quasiconformal homeomorphism between open subsets of Carnot groups are smooth with respect to the underlying Lie group structure [5]. Their approach

was through showing composition with 1-quasiconformal homeomorphisms preserve Q-harmonic functions. For open subsets of the boundaries of quaternionic hyperbolic space and the Cayley plane, Pansu showed that quasiconformal homeomorphisms are 1-quasiconformal, so combining this with Capogna–Cowling's theorem and results of Cowling–Ottazzi [6], even quasiconformal homeomorphisms between connected domains of the boundaries extend to unique isometries of the interior. One should compare this to how conformal mappings between connected domains of the n-sphere, for $n \geq 3$, extend to unique Möbius transformations.

Among a plethora of other quasi-isometric rigidity results, Bourdon and Pajot proved that a quasi-isometry of certain 2-dimensional hyperbolic buildings, specifically right-angled Fuchsian buildings, are within bounded distance of unique isometries [4]. This has been generalized by Xie to all Fuchsian buildings [25].

The results in this dissertation concern certain Loewner spaces with 1-dimensional Cheeger cotangent bundle. In this setting, quasiconformal mappings are automatically 1-quasiconformal. We show that composition with quasiconformal maps preserve Q-energy and Q-harmonic functions. Vaguely speaking, we show that Q-harmonic functions descend to graphs and for a certain class of spaces constructed in a self-similar manner, quasiconformal mappings are always local homotheties. We note that this phenomenon of Q-harmonic functions descending to graphs has not been observed in the literature before; this result about function theory is a new approach to rigidity of quasiconformal homeomorphisms.

1.2 Statement of results

We refer the reader to Section 2 for relevant definitions.

The objective of this paper is to analyze consequences of a preservation of Q-energy formula for inverse limit spaces. An inverse system of metric measure graphs

$$X_0 \overset{\pi_0}{\leftarrow} X_1 \overset{\pi_1}{\leftarrow} \cdots,$$

where $X_i = (X_i, d_i, \mu_i)$ is said to be *admissible* if it satisfies (A1)–(A7), Section 2.9, below. (Indices ranging over $\mathbb{Z}$ are also possible, but we prove our results for systems indexed over the nonnegative integers.) This condition is defined in [8] and implies that the resulting inverse limit space is a PI space.

First, we establish a preservation of Q-energy formula for bi-Lipschitz mappings between Ahlfors Q-regular limit spaces of admissible inverse systems (Theorem 3.3):

Theorem 1.1. *Let $f : U \subseteq X_\infty \to V \subseteq Y_\infty$ be a bi-Lipschitz homeomorphism between measurable subsets of Ahlfors Q-regular limit spaces of admissible inverse systems. Then, for any $u : V \to \mathbb{R}$ Lipschitz,*

$$\int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q = \int_V |Du|^Q \, d\mathcal{H}^Q.$$

The notation $|Du|$ is defined in Section 2.1. By a standard argument, this result extends to a quasiconformal homeomorphism between open subsets (Theorem 3.4):

Theorem 1.2. *Let X_∞, Y_∞ be Ahlfors Q-regular limit spaces of admissible inverse systems with $Q > 1$ and $f : U \subseteq X_\infty \to V \subseteq Y_\infty$ a quasiconformal homeomorphism between open subsets.*

For any $u : V \to \mathbb{R}$ Lipschitz, we have

$$\int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q = \int_V |Du|^Q \, d\mathcal{H}^Q.$$

Consequences of this theorem include preservation of Newtonian–Sobolev functions (Theorem 4.2) and Q-harmonic functions (Theorem 4.3), that is, if $f : U \to V$ is as in Theorem 3.4 and ρ_u denotes the minimal Q-weak upper gradient of u:

Theorem 1.3. *If $u \in N^{1,Q}(V)$ is bounded, then $u \circ f \in N^{1,Q}(U)$ and*

$$\|\rho_{u \circ f}\|_{L^Q(U)} = \|\rho_u\|_{L^Q(V)}.$$

Theorem 1.4. *If $u \in N^{1,Q}(V)$ is bounded and Q-harmonic, then $u \circ f \in N^{1,Q}(U)$ is Q-harmonic.*

We apply these results to a certain class of admissible inverse systems. The inverse system $X_\bullet$ is said to be in the class $\mathcal{F}$ if the conditions (B1)–(B3), Section 5.1, below hold for some fixed parameters $m, k \geq 2$. Roughly speaking, X_{i+1} is obtained in a self-similar construction by subdividing each edge of k copies of X_i into m parts, then gluing each newly constructed vertex to the $k-1$ corresponding newly constructed vertices in the other graphs.

Fix an inverse system $(X_\bullet, \pi_\bullet)$ in the class $\mathcal{F}$ with parameters $m, k \geq 2$. We first verify (Proposition 5.5) that μ_∞ is equal to $\mathcal{H}^Q$ with Q as in 5.1. We then show that X_∞ has isometries which are transitive over points which project to the same edge under the projection $X_\infty \to X_i$ and which are compactly supported in a quantitative sense (Proposition 5.6). Using uniqueness results for Q-harmonic functions (given boundary values), this is used to show that Q-harmonic functions descend to finite approximations in Section 5.2. Roughly speaking, for a Q-harmonic function f defined on an open set U and a point $x \in U$, in a neighborhood of x the value of f only depends on the projection to X_i, for i sufficiently large depending on the distance of x to the boundary of U, and the value of f will be piecewise affine with respect to this projection. Actually, Q'-harmonic functions descend in the same sense for any $Q' \in (1, \infty)$.

The idea for preservation of Q-energy and that Q-harmonic functions locally descend to finite graph approximations go back to unpublished joint work of M. Bonk and B. Kleiner. They proposed that the author of this dissertation work out the details for proving rigidity of bi-Lipschitz and quasiconformal homeomorphisms.

With $X_\bullet$ in the class $\mathcal{F}$ as above, using that the height on X_∞ (the projection onto X_0, which is a unit interval) is Q-harmonic, that f preserves Q-harmonic functions, and applying uniqueness results (given boundary values) for Q-harmonic functions, in Section 5.3 we show that quasiconformal homeomorphisms between open subsets of X_∞ are local homotheties, since mappings which are affine on height are local homotheties.

The projection $\Pi_i : X_\infty \to X_i$ is defined to be the inverse limit of the projections $\pi_i \circ \cdots \circ \pi_{j-2} \circ \pi_{j-1} : X_j \to X_i$. An edge preimage subspace of level i, shortened to *edge preimage*, is a preimage in X_∞ of an edge of the graph X_i under the projection Π_i. In Section 5.3, we prove:

Theorem 1.5. *Let $f : U \to V$ be a quasiconformal homeomorphism between connected open subsets of X_∞. Then there exists $\ell \in \mathbb{Z}$ such that f is locally an $m^{-\ell}$-homothety. In particular, for each $x \in U$, there exists i and $E \subseteq U$ with $E = \bigcup_{r=1}^{N} \Pi_i^{-1}(e_i^r)$, where e_i^r are the edges in X_i intersecting $\Pi_i(x)$, such that the restriction of f to each $\Pi_i^{-1}(e_i^r)$ is an $m^{-\ell}$-homothety between $\Pi_i^{-1}(e_i^r)$ and an edge preimage of level $i + \ell$.*

Theorem 1.6. *Let $f : X_\infty \to X_\infty$ be a quasiconformal homeomorphism. Then f is an isometry. Moreover, there exists i such that the restriction of f to any edge preimage of level i is an isometry onto an edge preimage of level i.*

Let $S \subseteq X_\infty$ denotes the points of height 0 or 1.

Theorem 1.7. *Let $f : X_\infty \setminus S \to X_\infty$ be an open quasiconformal embedding.*

(1) If $m \geq 3$, then f is an m^{-i}-homothety onto some edge preimage of level i.

(2) If $m = 2$ and the conclusion of (1) does not hold, then f is still an m^{-i}-homothety in this case. The restriction of f to any edge preimage of level 1 is an m^{-i}-homothety onto some edge preimage of level $1 + i$.

Note, in a preprint from 2001 ([17]), T. Laakso states that he can use tangent cones to prove that bi-Lipschitz and quasiconformal maps between open subsets of Laakso spaces are local homotheties for almost every dimension $Q > 1$, though to the author of this dissertation's knowledge such a result has not appeared in print.

In Section 6, we discuss applications of these ideas to the boundaries of certain 2-dimensional hyperbolic buildings; in particular we prove that quasiconformal homeomorphisms between open

subsets preserve Q-energy and Q-harmonic functions. We also give conditions which imply that Q-harmonic functions descend to graphs, specifically tree-walls.

2 Preliminaries

2.1 Differentiability spaces

For background on differentiability spaces, we refer the reader to [9] or [15]. We summarize the required definitions.

Fix a metric measure space (X, d, μ). Define

$$\operatorname{var}_{x,r} f = \sup_{y \in B(x,r)} d(f(x), f(y))/r$$

for $x \in X$ and $r > 0$. Let

$$L_{x,r} f = \sup_{0 \le s \le r} \operatorname{var}_{x,s} f$$

and

$$\operatorname{Lip}_x f = \lim_{r \to 0} L_{x,r} f = \limsup_{r \to 0} \operatorname{var}_{x,r} f.$$

An N-dimensional *differentiability chart* $\varphi : U \subseteq X \to \mathbb{R}^N$ is a Lipschitz mapping with U measurable such that, for all Lipschitz functions $u : U \to \mathbb{R}$ we can define for μ-a.e. $x \in X$ a unique differential $Du(x) = (D_1 u(x), \ldots, D_N u(x))$ such that

$$u(y) - u(x) = \sum_{i=1}^{N} D_i u(x)(\varphi_i(y) - \varphi_i(x)) + o(d(y, x))$$

as $y \to x$.

X is said to be a *differentiability space* if is is given the structure of a countable collection of differentiability charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^{N_\alpha}$, where the sets U_α cover X up to a μ-null set. This collection of charts is called a *differentiable structure*.

Note: If U_α and U_β intersect in a set of positive measure, then $N_\alpha = N_\beta$. Also, any measurable subset of a differentiablility space admits the structure of a differentiability space in a natural way.

The differentiable structure allows us to define a measurable cotangent bundle T^*X whose fibers are defined μ-a.e. Given a Lipchitz function u defined on X, we can define Du to be a section of this cotangent bundle defined μ-a.e. T^*X comes equipped with a Finsler metric $|\cdot|$ such that, for μ-a.e. x,

$$|Du(x)| = \operatorname{Lip}_x u.$$

If Y is another differentiability space , given a bi-Lipschitz homeomorphism $f : X \to Y$ such that the pushforward of the measure on X is absolutely continuous with respect to the measure on Y, there is an *induced map* $f_x^* = f^* : T^*_{f(x)}Y \to T^*_x X$ for a.e. $x \in X$ (see the discussion after Remark 4.44 in [7]) with the following properties: For Lipschitz $u : Y \to \mathbb{R}$, for a.e. $x \in X$, we have

$$f^*(Du(f(x))) = D(u \circ f)(x).$$

In the definition of differentiablility space, if all the N_α equal 1, we say that X has *1-dimensional cotangent bundle*. When X and Y both have 1-dimensional cotangent bundle, we define $|Df(x)|$ to be the operator norm of f^*. Since f^* is a linear map between 1-dimensional normed spaces,

$$|D(u \circ f)(x)| = |f^*(Du(f(x)))| = |Du(f(x))| \cdot |Df(x)|.$$

If Z is a differentiablility space with 1-dimensional cotangent bundle, $g : Y \to Z$ is bi-Lipschitz, and the pushforward of the measure on Y is absolutely continuous with respect to the measure on Z, then

$$|D(g \circ f)(x)| = |Dg(f(x))| \cdot |Df(x)|,$$

for a.e. $x \in X$. Hence, if the pushforward of the measure on Y under f^{-1} is absolutely continuous

with respect to the measure on X,

$$|Df^{-1}(y)| = 1/|Df(f^{-1}(y))| \tag{2.1}$$

for a.e. $y \in Y$ (it is straighforward to check that $|Df^{-1}(y)|$ is positive for a.e. $y \in Y$).

2.2 Modulus of curve families

Let (X, μ) be a metric measure space and $Q \in [1, \infty)$. If Γ is a collection of paths in X (to be specific, the domains of the paths in Γ can be open, closed, half-open, bounded, or unbounded; typically, we consider curve families joining bounded sets, in which case the domains of the paths will be closed), the *Q-modulus* of Γ is defined to be

$$\operatorname{Mod}_Q(\Gamma) = \inf \|\rho\|^Q_{L^Q(X)},$$

where the infimum is taken over all nonnegative Borel measurable $\rho : X \to [0, \infty]$ such that $\int_\gamma \rho \geq 1$ for all locally rectifiable $\gamma \in \Gamma$. If Γ does not contain any locally rectifiable curves, then define $\operatorname{Mod}_Q(\Gamma) = 0$. Note that Mod_Q is an outer measure on the set of all paths.

A property defined on a collection of paths is said to hold for *Q-almost every path*, or Q-a.e. path, in that collection if the set of paths in that collection it does not hold for has Q-modulus zero.

2.3 Upper gradients and weak upper gradients

If u is a real-valued function on X, a nonnegative Borel measurable function ρ is said to be an *upper gradient* of u (or a strong upper gradient, to contrast with weak upper gradients below) if,

for all closed rectifiable paths γ in X,

$$|u(x) - u(y)| \leq \int_\gamma \rho,$$

where x and y are the endpoints of γ. If the above inequality holds for Q-a.e. closed rectifiable path, we say that ρ is a *Q-weak upper gradient* of u.

2.4 Poincaré inequalities

Let (X, μ) be a metric measure space and $Q \in [1, \infty)$. We say that X admits a *weak* $(1, Q)$-*Poincaré inequality* (the "weak" is often dropped) if there is a constant $C \geq 1$ such that

$$\fint_B |u - u_B| \, d\mu \leq C \cdot \operatorname{diam} B \cdot \left(\fint_{CB} \rho^Q \right)^{1/Q} \tag{2.2}$$

for all balls $B = B(x, r)$ in X, all bounded continuous functions u defined in X, and all upper gradients ρ of u in B. We define $\fint_B u \, d\mu = \frac{1}{\mu(B)} \int u \, d\mu$ and $CB = B(x, Cr)$. If (X, μ) satisfies 2.2 with CB replaced by B, we say that X admits a *strong* $(1, Q)$-*Poincaré inequality*. If (X, μ) admits a $(1, Q)$-Poincaré inequality for some $Q \in [1, \infty)$ and μ is doubling, we say that X is a *PI space*.

The following theorem of Cheeger ([7], Theorem 4.38) states that PI spaces admit a differentiable structure. (See also [15] for an exposition.)

Theorem 2.1 (Cheeger differentiable structure)**.** *If (X, μ) is a PI space, then X admits a differentiable structure such that the dimension of the charts is bounded above by a constant depending only on the constant C in the Poincaré inequality and the doubling constant of μ.*

While we do not need the following result as the spaces we consider in this paper admit $(1, 1)$-Poincaré inequalitites, note that the $(1, Q)$-Poincaré inequality can be strengthened by the following theorem ([12], Theorem 1.0.1) of Keith and Zhong:

Theorem 2.2 (Keith–Zhong self-improvement)**.** *Let X be a complete PI space that admits a $(1, Q)$-Poincaré inequality with $Q > 1$. Then there exists an $\varepsilon > 0$ with $1 \leq Q - \varepsilon < Q$ such that X admits a $(1, Q')$-Poincaré inequality for all $Q' > Q - \varepsilon$ (quantitative).*

2.5 Loewner spaces

The Loewner condition is a property that is related to the Poincaré inequality. We define this here because some theorems we cite are stated for Loewner spaces. Suppose that (X, μ) is a rectifiably-connected metric measure space and fix $Q > 1$. Define

$$\phi(t) = \inf \{\mathrm{Mod}_Q(\Gamma_{E,F}) : \Delta(E, F) \leq t\}, \tag{2.3}$$

where the infimum is over pairs (E, F) of disjoint *continuums* (compact, connected sets in X containing at least two points), $\Gamma_{E,F}$ is the set of closed paths joining E to F, and

$$\Delta(E, F) = \frac{\mathrm{dist}(E, F)}{\min(\mathrm{diam}\, E, \mathrm{diam}\, F)}$$

(in 2.3, an infimum over an empty set is understood to be zero). X is said to be a *Loewner space of exponent Q* or a *Q-Loewner space* if $\phi(t) > 0$ for all $t > 0$. If X is an Ahlfors Q-regular Q-Loewner space we just write that X is an Ahlfors Q-regular Loewner space.

A metric space (X, d) is said to be *quasiconvex* if there is some constant $C \geq 1$ such that any two points x, y can be joined by some path of length at most $Cd(x, y)$. We say that (X, d) is *linearly locally connected* if there is some constant $C \geq 1$ such that, for all $r > 0$, (1) any two points in $B(x, r)$ can be joined by a continuum in $B(x, Cr)$ and (2) any two points in $X \setminus \bar{B}(x, r)$ can be joined by a continuum in $X \setminus \bar{B}(x, r/C)$. The following is Theorem 3.13 in [11]:

Theorem 2.3. *Let X be an Ahlfors Q-regular Loewner space with $Q > 1$. Then X is quasiconvex and linearly locally connected (quantitative).*

By Theorem 17.1 in [7] (which is an exposition of an argument of Stephen Semmes), a complete PI space is quasiconvex. This allows us to drop the hypothesis of "φ-convexity" in Corollary 5.13 of [11], so we state it in this form:

Theorem 2.4. *Let X be a proper Ahlfors Q-regular metric space (equivalently, a complete Ahlfors Q-regular metric space) with $Q > 1$. Then X is a Q-Loewner space if and only if X admits a $(1, Q)$-Poincaré inequality (quantitative).*

2.6 Quasiconformal and quasisymmetric maps

Let $f : X \to Y$ be a homeomorphism between metric spaces. Given a constant $H \geq 1$, we say that f is *H-quasiconformal* if

$$\limsup_{r \to 0} \frac{\sup \{d(f(x), f(y)) : d(x, y) \leq r\}}{\inf \{d(f(x), f(y)) : d(x, y) \geq r\}} \leq H.$$

Given an increasing homeomorphism $\eta : [0, \infty) \to [0, \infty)$, we say that f is *η-quasisymmetric* if

$$\frac{d(f(x), f(y))}{d(f(x), f(z))} \leq \eta \left(\frac{d(x, y)}{d(x, z)} \right)$$

for all triples of distinct points x, y, z in X. Is is clear that, if f is quasisymmetric, then f is quasiconformal. In the other direction, for homeomorphisms between Euclidean domains of dimension > 1, it was first proved by Väisälä in [24] that quasiconformal homeomorphisms are locally quasisymmetric.

The quasiconformal condition is easier to check and the quasisymmetry condition is easier to use, so one would like to be able to interchange between the two. The paper [11] shows that Ahlfors Q-regular Loewner spaces are a natural generality to study the geometry of quasiconformal maps. In particular, a local-to-global principle also holds in this setting (this is Theorem 4.7 in [11]):

Theorem 2.5. *Suppose that X, Y are Ahlfors Q-regular spaces with $Q > 1$, X is Q-Loewner,*

and Y is linearly locally connected. If $f : X \to Y$ is an H-quasiconformal homeomorphism, then each point of x has a neighborhood U in which the restriction of f is an η-quasisymmetric homeomorphism onto its image; η only depends on H, X, and Y. Moreover, U can be taken to be equal to $B(x, r)$ if $Y \setminus \bar{B}(f(x), L_{x,2r}f)$ is nonempty.

See also [23], Theorem 10.9, for a local version.

2.7 Newtonian spaces, minimal weak upper gradients, and Q-harmonic functions

Cheeger defines a version of Sobolev spaces on metric measure spaces in [7] and studies Q-harmonic functions. Later, Shanmugalingam introduces Newtonian spaces, also known as Newtonian–Sobolev spaces, in [21], which she has verified give rise to the same spaces as Cheeger's provided that $Q > 1$. We work with Shanmugalingam's definition. See also [1] and [22] for the relevant background. Note that Shanmugalingam states results for the strong form of the Poincaré inequality, but it is commented after Definition 2.9 in [22] that this is done to simplify notation and the results also hold for the weak form of the Poincaré inequality.

Fix a metric measure space X. We denote the functions that have a Q-integrable Q-weak upper gradient by $\tilde{N}^{1,Q}(X)$, and define

$$\|u\|_{N^{1,Q}(X)} = \|u\|_{L^Q(X)} + \inf \|\rho\|_{L^Q(X)},$$

where the infimum is over all Q-integrable Q-weak upper gradients ρ of u. Identifying u and v if $\|u - v\|_{N^{1,Q}(X)} = 0$, we obtain a Banach space, the *Newtonian space* $N^{1,Q}(X)$.

A function $u : X \to \mathbb{R}$ is said to be ACC_Q (absolutely continuous on Q-a.e. curve) if $u \circ \gamma$ is absolutely continuous for Q-a.e. closed rectifiable curve $\gamma : [0, \mathrm{len}(\gamma)] \to X$ parameterized by arc-length. Any function $u \in \tilde{N}^{1,Q}(X)$ is ACC_Q. Moreover, such a u has a *minimal Q-integrable*

weak upper gradient ρ_u in the sense that ρ_u has a.e. values $\leq$ those of any other weak upper gradient of u. If X is a PI space that admits a $(1, Q)$-Poincaré inequality and u is Lipschitz, then $\rho_u = \mathrm{Lip}_x u$ for a.e. $x \in X$.

We need the following lemma concerning ACC_Q functions:

Lemma 2.6. *Let $u : X \to \mathbb{R}$ be ACC_Q on a metric measure space X. Suppose that $U \subseteq X$ is an open set. If $u \mid \partial U \equiv 0$, then the function $\tilde{u}$ defined by $\tilde{u} \equiv u$ in U and $\tilde{u} \equiv 0$ in $X \setminus U$ is ACC_Q.*

Proof. For Q-a.e. rectifiable $\gamma : [0, \mathrm{len}(\gamma)] \to X$ parameterized by arc-length, we know that $u \circ \gamma$ is absolutely continuous. Pick such a γ. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that, for any positive integer N and a finite sequence of pairwise disjoint subintervals $(x_1, y_1), \ldots, (x_N, y_N)$ of $[0, \mathrm{len}(\gamma)]$ with $\sum |x_i - y_i| < \delta$, we have $\sum |u \circ \gamma(x_i) - u \circ \gamma(y_i)| < \varepsilon$.

With δ as above, suppose that we are given a positive integer N and a finite sequence of pairwise disjoint subintervals $(x_1, y_1), \ldots, (x_N, y_N)$ of $[0, \mathrm{len}(\gamma)]$ with $\sum |x_i - y_i| < \delta$. Let $I_1 \subseteq \{1, \ldots, N\}$ denote those indices i such that $x_i, y_i \in U$, and let I_2 denote those indices i such that one of x_i, y_i is in U and the other is in $X \setminus U$.

If $i \notin I_1 \cup I_2$, then $|\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| = 0$. If $i \in I_1$, then $|\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| = |u \circ \gamma(x_i) - u \circ \gamma(y_i)|$. If $i \in I_2$, then there exists $t_i \in [x_i, y_i]$ with $\gamma(t_i) \in \partial U$. If $x_i \in U$ (resp. $y_i \in U$) let $(\tilde{x}_i, \tilde{y}_i) = (x_i, t_i)$ (resp. $(\tilde{x}_i, \tilde{y}_i) = (t_i, y_i)$). Then observe that $|\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| \leq |u \circ \gamma(\tilde{x}_i) - u \circ \gamma(\tilde{y}_i)|$. In any case,

$$\begin{aligned}
\sum_{i=1}^{N} |\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| &= \sum_{i \in I_1} |\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| + \sum_{i \in I_2} |\tilde{u} \circ \gamma(x_i) - \tilde{u} \circ \gamma(y_i)| \\
&\leq \sum_{i \in I_1} |u \circ \gamma(x_i) - u \circ \gamma(y_i)| + \sum_{i \in I_2} |u \circ \gamma(\tilde{x}_i) - u \circ \gamma(\tilde{y}_i)| \\
&< \varepsilon
\end{aligned}$$

because $\sum_{i \in I_1} |x_i - y_i| + \sum_{i \in I_2} |\tilde{x}_i - \tilde{y}_i| \leq \sum_{i=1}^{N} |x_i - y_i| < \delta$. □

The Q-capacity of a set $E \subseteq X$ is defined to be $\inf \|u\|^Q_{N^{1,Q}(X)}$, where the infimum is taken over the functions in $N^{1,Q}(X)$ that have a representative which is ≥ 1 on E.

For $E \subseteq X$, we define the *Newtonian space of functions that vanish outside* E. Let $\tilde{N}^{1,Q}_0$ denote the functions $u : E \to [-\infty, \infty]$ such that there exists $\tilde{u} \in \tilde{N}^{1,Q}$ with the Q-capacity of $\{x \in X \setminus E : \tilde{u}(x) \neq 0\}$ equal to 0. Then

$$\|u\|_{N^{1,Q}_0(E)} = \|\tilde{u}\|_{N^{1,Q}(X)}$$

does not depend on the choice of $\tilde{u}$. Identifying u and v if $\|u - v\|_{N^{1,Q}_0(E)} = 0$, we obtain a Banach space $N^{1,Q}_0(E)$.

We say that $u \in N^{1,Q}(X)$ is *Q-harmonic* in $E \subseteq X$ if for all $v \in u + N^{1,Q}_0(E)$ it holds that

$$\|\rho_u\|^Q_{L^Q(X)} \leq \|\rho_v\|^Q_{L^Q(X)}.$$

If u is Q-harmonic in X we just say that u is Q-harmonic.

Suppose that X is a complete PI space supporting a $(1, Q)$-Poincaré inequality. Then (1) Lipschitz functions are dense in $N^{1,Q}(X)$; (2) if E is an open subset of X, then $N^{1,Q}_0(E)$ is the closure of the Lipschitz functions with compact support in E under the $N^{1,Q}$ norm (so we can define $N^{1,Q}_0(E)$ without reference to X globally); and (3) If E is an open subset of X, then $N^{1,Q}(E)$ is the closure of the locally Lipschitz functions in E with finite $N^{1,Q}(E)$ energy (see Theorem 5.47 in [1]).

We have the following existence, uniqueness, and regularity theorem from [22], Theorem 5.6 and Remark 6.1 (note that [22], Remark 6.1, references [13]):

Theorem 2.7. *Let X be a complete PI space supporting a $(1, Q)$-Poincaré inequality with $Q > 1$. If U is a bounded open subset of X with the interior of $X \setminus U$ nonempty, then for any $w \in N^{1,Q}(X)$ there exists $u \in N^{1,Q}(X)$ that is Q-harmonic in U such that $u - w \in N^{1,Q}_0(U)$. If U is path-connected, then u is unique up to a set of Q-capacity zero.*

If, in addition, X satisfies a $(1, Q')$-Poincaré inequality for some $1 \leq Q' < Q$ (this is automatically true by Keith–Zhong self-improvement), then u has a representative that is Hölder continuous in U.

We want to phrase in a different way how Q-harmonic functions in PI spaces are determined by their boundary values, but we first need a lemma:

Lemma 2.8. *Let X be a metric measure space. Suppose that $U \subseteq X$ is an open subset with $\partial U \neq \emptyset$. If $u, v \in \tilde{N}^{1,Q}(X)$ are such that $u = v$ in ∂U, then the function $\tilde{u}$ defined by $\tilde{u} = u$ in $X \setminus U$ and $\tilde{u} = v$ in U is in $\tilde{N}^{1,Q}(X)$.*

Proof. Let w be a function equal to $v - u$ in U and 0 outside U. Since functions in $N^{1,Q}(X)$ are ACC_Q, it follows that w is ACC_Q by Lemma 2.6. Then $w \in N^{1,Q}(X)$ by [22], Lemma 3.2. Hence $\tilde{u} = u + w$ is in $N^{1,Q}(X)$. □

We can then rephrase the uniqueness for Q-harmonic functions:

Lemma 2.9. *Let X be a complete PI space supporting a $(1, Q)$-Poincaré inequality with $Q > 1$. Let $u, v \in N^{1,Q}(X)$. Suppose that $U \subseteq X$ is a path-connected open subset with the interior $X \setminus U$ nonempty. Suppose that u, v are Q-harmonic in U. If $u = v$ in ∂U, then $u = v$ in U (up to a set of Q-capacity zero).*

Proof. Consider the function $\tilde{u}$ equal to u in $X \setminus U$ and v in U. By the previous lemma, $\tilde{u} \in N^{1,Q}(X)$. The result follows from Theorem 2.7 since then $\tilde{u} = u$ up to a set of Q-capacity zero. □

2.8 Blowups

We refer the reader to Definition 8.58 in [9]. Given a sequence of scale factors $r_i \to 0$, a *blowup* of a pointed metric space (X, d, x) (where d is the distance function and $x \in X$) along the scales r_i is an ultralimit of the pointed metric spaces $(X, \frac{1}{r_i}d, x) = (\frac{1}{r_i}X, x)$, denoted $(\hat{X}, \hat{d}, \hat{x})$ or $(\hat{X}, \hat{x})$.

Given suitable geometric constraints on a mapping $f : X \to Y$ (e.g., Lipschtiz), a *blowup* of f at $x \in X$ along the scales r_i is an ultralimit of the mappings between $(\frac{1}{r_i}X, x) \to (\frac{1}{r_i}Y, f(x))$ equal to f set-wise. When the scale factors and non-principle ultrafilter are fixed, blowups of a metric space at a point are unique and blowups of mappings at a point are unique when they are defined.

2.9 Admissible inverse systems

We reference [8] and [9], Section 9. Consider an inverse system of metric measure graphs

$$\cdots \overset{\pi_{i-1}}{\leftarrow} X_i \overset{\pi_i}{\leftarrow} X_{i+1} \overset{\pi_{i+1}}{\leftarrow} \cdots,$$

where $X_i = (X_i, d_i, \mu_i)$ and i ranges over either $\mathbb{Z}$ (in which case the inverse system is called *signed*) or the nonnegative integers (in which case the inverse system is called *unsigned*). This inverse system is said to be *admissible* if for some fixed integer $m \geq 2$ and constants $\Delta, \theta, C \in (0, \infty)$ the following holds:

(A1) (X_i, d_i) is a nonempty connected metric graph with all vertices of degree $\leq \Delta$ and edges isometric to an interval of length m^{-i}

(A2) If X_i' denotes the graph obtained by subdividing each edge of X_i into m edges of length $m^{-(i+1)}$, then π_i induces a map $X_{i+1} \to X_i'$ which is open, simplicial, and isometric on every edge

(A3) For all $x_i \in X_i'$, each preimage $\pi_i^{-1}(\{x_i\}) \subseteq X_{i+1}$ has diameter $\leq \theta m^{-(i+1)}$

(A4) μ_i restricts to a multiple of arclength on each edge of X_i, and if e_1, e_2 are two adjacent edges in X_i we have $\mu_i(e_1)/\mu_i(e_2) \in [C^{-1}, C]$

(A5) The pushforward $(\pi_i)_*\mu_{i+1}$ is equal to μ_i

(A6) For all vertices $v_i \in X_i'$ and $v_{i+1} \in \pi_i^{-1}(v_i)$,

$$\frac{\mu_{i+1}(\pi_i^{-1}(e_i) \cap \operatorname{star}(v_{i+1}, X_{i+1}))}{\mu_i(e_i)}$$

is the same for all edges $e_i \subseteq \operatorname{star}(v_i, X_i')$. Here $\operatorname{star}(v, G)$ denotes the union of the edges containing v in a graph G

(A7) If the system $\{(X_\bullet, \pi_\bullet)\}$ is unsigned, assume (X_0, μ_0) is isomorphic to the unit interval with length measure and denote

$$\varphi_i = \pi_0 \circ \cdots \circ \pi_{i-1}.$$

If $\{(X_\bullet, \pi_\bullet)\}$ is signed, assume the existence of open surjective maps $\varphi_i : X_i \to \mathbb{R}$ which (viewing $\mathbb{R}$ as a graph with edges $\{[km^{-i}, (k+1)m^{-i})] : k \in \mathbb{Z}\}$) are simplicial, restrict to isometries on edges, and satisfy $\varphi_i \circ \pi_i = \varphi_{i+1}$

The main result of [8] is that the measured Gromov–Hausdorff limit $(X_\infty, d_\infty, \mu_\infty)$ of an admissible inverse system (X_i, d_i, μ_i) is a PI spaces admitting a $(1,1)$-Poincare inequality. Moreover, the inverse limit $\varphi = \varphi_\infty$ of the functions φ_i in (7) serves as a global 1-dimensional differentiability chart in X_∞ with $\operatorname{Lip}_\bullet \varphi \equiv 1$. We call φ (resp. φ_i) the *height* in X_∞ (resp. X_i).

A path γ parameterized by arclength with values in X_∞ is said to be *monotone* if $|\varphi(\gamma(s)) - \varphi(\gamma(t))| = |s - t|$ for all s, t in the domain of γ. Notice that the length of a monotone path is the length of its domain.

We define $\Pi_i^{i+j} : X_{i+j} \to X_i$ by

$$\Pi_i^{i+j} = \pi_{i+j-1} \circ \cdots \circ \pi_{i+1} \circ \pi_i \tag{2.4}$$

and $\Pi_i : X_\infty \to X_i$ by

$$\Pi_i = \lim_{j \to \infty} \Pi_i^{i+j} \tag{2.5}$$

in the sense of inverse limit of mappings.

The measure μ_∞ admits an *Alberti representation* by [9], Theorem 9.13(2) (for simplicity, we state this for unsigned $X_\bullet$): Identify X_0 with $[0,1]$. Let Γ denote the set of monotone geodesics $\gamma : [0,1] \to X_\infty$, where $\varphi_\infty \circ \gamma = \mathrm{id}_{[0,1]}$. There is a measure M on Γ such that, for all integrable $u : X_\infty \to \mathbb{R}$, we have

$$\int_{X_\infty} u \, d\mu_\infty = \int_\Gamma \left(\int_0^1 u \circ \gamma(t) \, dt \right) dM. \tag{2.6}$$

In Section 3.1, we need the following lemma:

Lemma 2.10. *Let X_∞, Y_∞ be limit spaces of admissible inverse systems $(X_\bullet, \pi_\bullet)$, $(Y_\bullet, \tilde{\pi}_\bullet)$. Denote the height functions on X_∞, Y_∞ by φ, ψ. If a Lipschitz mapping $f : X_\infty \to Y_\infty$ preserves height up to a constant L, i.e., $\psi(f(x)) - \psi(f(y)) = L \cdot (\varphi(x) - \varphi(y))$ for all $x, y \in X_\infty$, then f is $|L|$-Lipschitz.*

Proof. Let $\varepsilon > 0$ and $x, y \in X_\infty$. For i sufficiently large, $\operatorname{diam} \Pi_i^{-1}(\Pi_i(y)) \le \varepsilon$, by (3). Since X_∞ is the inverse limit of $X_\bullet$, for i sufficiently large, $|d_{X_i}(\Pi_i(x), \Pi_i(y)) - d_{X_\infty}(x, y)| \le \varepsilon$.

For such an i which satisfies the conditions in the previous paragraph, let γ_i be a piecewise-monotone geodesic in X_i joining $\Pi_i(x)$ to $\Pi_i(y)$. By (2), we can lift γ_i to a path in X_{i+1}, then to a path in X_{i+2}, ..., and take an inverse limit to obtain a piecewise-monotone path γ in X_∞ joining x to a point in $\Pi_i^{-1}(\Pi_i(y))$. Notice that γ has the same length as γ_i. Since f preserves height up to the constant L, using that γ is piece-wise monotone we see that

$$\begin{aligned} \operatorname{length}(f \circ \gamma) &= |L| \cdot \operatorname{length}(\gamma) \\ &= \operatorname{length}(\gamma_i) \\ &= |L| \cdot d_{X_i}(\Pi_i(x), \Pi_i(y)) \\ &\le |L| \cdot d_{X_\infty}(x, y) + |L|\varepsilon. \end{aligned}$$

Since γ joins x to a point of distance $\leq \varepsilon$ from y, we see that

$$d_{Y_\infty}(f(x), f(y)) \leq |L| \cdot d_X(x, y) + |L|\varepsilon + C\varepsilon,$$

where C is the global Lipschitz constant of f. □

We also need in Section 5.2 that the height function on a limit space of an admissible inverse system is Q-harmonic. We state this for unsigned systems.

Lemma 2.11. *Let X_∞ be a limit space of an unsigned admissible inverse system with height function $\varphi : X_\infty \to [0, 1]$. Then φ is Q-harmonic in $U = X_\infty \setminus (\varphi^{-1}(0) \cup \varphi^{-1}(1))$.*

Proof. Suppose that v is a Lipschitz function with compact support in U. Let N denote the points of U at which $\varphi + v$ is not differentiable; then N is μ_∞-null.

Let Γ and M be as in 2.6. Then M-a.e. $\gamma \in \Gamma$ has image that intersects N in an $\mathcal{H}^1$-null set. Let $\tilde{\Gamma} \subseteq \Gamma$ denote this full measure subset and consider such $\gamma \in \tilde{\Gamma}$.

Observe that $|(\varphi + v) \circ \gamma)'(t)|$ exists for $\mathcal{H}^1$-a.e. $t \in [0, 1]$. It is known that the linear function $t \mapsto t$ on $[0, 1]$ minimizes the Q-energy $\int_0^1 |u'(t)|^Q \, dt$ amongst all Lipschitz $u : [0, 1] \to \mathbb{R}$ with $u(0) = 0$ and $u(1) = 1$. Then

$$\begin{aligned}\int_0^1 |(\varphi + v) \circ \gamma)'(t)|^Q \, dt &\geq \int_0^1 |(\varphi \circ \gamma)'(t)|^Q \\ &= \int_0^1 1 \, dt \\ &= \int_0^1 (\mathrm{Lip}_{\gamma(t)} \varphi)^Q \, dt,\end{aligned}$$

since $\mathrm{Lip}_\bullet \varphi \equiv 1$.

If $\gamma(t)$ is a point of differentiability of $\varphi + v$, then $((\varphi + v) \circ \gamma)'(t)$ exists and $|((\varphi + v) \circ \gamma)'(t)| =$

$\mathrm{Lip}_{\gamma(t)}(\varphi + v)$. This holds for $\mathcal{H}^1$-a.e. $t \in [0,1]$, so we obtain

$$\begin{aligned}\int_0^1 \mathrm{Lip}_{\gamma(t)}(\varphi+v)^Q \, dt &= \int_0^1 |(\varphi+v)\circ\gamma)'(t)|^Q \, dt \\ &\geq \int_0^1 (\mathrm{Lip}_{\gamma(t)}\, \varphi)^Q \, dt.\end{aligned}$$

Thus,

$$\begin{aligned}\int_{X_\infty} \mathrm{Lip}_\bullet(\varphi+v)^Q \, d\mu_\infty &= \int_{\gamma\in\tilde{\Gamma}} \left(\int_0^1 \mathrm{Lip}_{\gamma(t)}(\varphi+v)^Q \, dt \right) dM \\ &\geq \int_{\gamma\in\tilde{\Gamma}} \left(\int_0^1 (\mathrm{Lip}_{\gamma(t)}\, \varphi)^Q \, dt \right) dM \\ &= \int_{X_\infty} (\mathrm{Lip}_\bullet\, \varphi)^Q \, d\mu_\infty,\end{aligned}$$

as desired. □

3 Preservation of Q-energy

3.1 Bi-Lipschitz homeomorphisms

We could not find a proof of the following lemma in the literature, though it probably has been used before. It is just an infinitesimal version of the fact that a C-Lipschitz map scales Q-dimensional Hausdorff measure of sets by at most C^Q. Note that, for simplicity, we abuse notation and use $\mathcal{H}^Q$ to denote the Q-dimensional Hausdorff measure on any metric space.

Lemma 3.1. *Let X and Y be Ahlfors Q-regular metric spaces and $f : X \to Y$ a bi-Lipschitz homeomorphism. Then, for a.e. $y \in Y$, the pushforward of the Q-dimensional Hausdorff measure satisifies*

$$\frac{df_*\mathcal{H}^Q}{d\mathcal{H}^Q}(y) \leq (\operatorname{Lip}_y f^{-1})^Q.$$

Proof. The Radon–Nikodym derivative of $f_*\mathcal{H}^Q$ with respect to $\mathcal{H}^Q$ is

$$\lim_{r\to 0} \frac{\mathcal{H}^Q(f^{-1}B(y,r))}{\mathcal{H}^Q(B(y,r))}$$

for a.e. $y \in Y$. Choose a sequence $r_i \to 0$. Note that $L_{y,r_i}f^{-1} \to \operatorname{Lip}_y f^{-1} = F(y)$ for a.e. $y \in Y$. Choose y to be such a point that is also a point of approximate continuity of F. There exists a set $E \ni y$ of Lebesgue density 1 at y such that the restriction of F to E is continuous at y.

Let $\varepsilon > 0$. If r is sufficiently small, then (noting that $F(y)$ is positive) $F \leq (1+\varepsilon)F(y)$ in $B(y,r) \cap E$ and $\mathcal{H}^Q(E^c \cap B(y,r)) \leq \varepsilon \cdot \mathcal{H}^Q B(y,r)$. Fix such an r, and then by Egorov's theorem there exists $E' \subseteq E$ on which the convergence of $L_{y,r_i}f^{-1}$ to F is uniform and such that $\mathcal{H}^Q(E'^c \cap E) \leq \varepsilon \cdot \mathcal{H}^Q B(y,r)$.

Since the convergence of $L_{y,r_i}f^{-1}$ to F is uniform on E', it follows that each point of $E' \cap B(y,r)$ has a neighborhood in which the restriction of f^{-1} to this neighborhood is $(1+2\varepsilon)F(y)$-Lipschitz,

and so $\mathcal{H}^Q f^{-1}(E' \cap B(y,r)) \leq (1+2\varepsilon)^Q F(y)^Q \mathcal{H}^Q(E' \cap B(y,r))$. Now,

$$\begin{aligned}\mathcal{H}^Q f^{-1} B(y,r) &= \mathcal{H}^Q f^{-1}(E' \cap B(y,r)) + \mathcal{H}^Q f^{-1}(E'^c \cap E \cap B(y,r)) + +\mathcal{H}^Q f^{-1}(E^c \cap B(y,r)) \\ &\leq (1+2\varepsilon)^Q F(y)^Q \mathcal{H}^Q(E' \cap B(y,r)) + C^Q \varepsilon \mathcal{H}^Q B(y,r) + C^Q \varepsilon \mathcal{H}^Q B(y,r) \\ &\leq (1+2\varepsilon)^Q F(y)^Q \mathcal{H}^Q B(y,r) + C^Q \varepsilon \mathcal{H}^Q B(y,r) + C^Q \varepsilon \mathcal{H}^Q B(y,r),\end{aligned}$$

where C is the global Lipschitz constant of f^{-1}. The lemma follows. □

Lemma 3.2. *Let $U \subseteq X_\infty$ and $V \subseteq Y_\infty$ be measurable subsets of Ahlfors Q-regular limit spaces of admissible inverse systems. If $f : U \to V$ is a bi-Lipschitz homeomorphism, then for a.e. $x \in U$, we have* $\operatorname{Lip}_x f = |Df(x)|$.

Proof. Let φ and ψ denote the height functions on X_∞ and Y_∞.

Let x be a point of differentiability of $\psi \circ f$. Fix a sequence of scales for which we can define blowups. Since

$$\psi \circ f, \qquad \psi \circ f(x) + D(\psi \circ f)(x) \cdot (\varphi - \varphi(x))$$

agree up to $o(d(\,\cdot\,, x))$, their blowups at x agree. Denote this blowup by $\widehat{\psi \circ f}$ and also the blowups of f at x and ψ at $f(x)$ by $\hat{f}$ and $\hat{\psi}$, so that $\widehat{\psi \circ f} = \hat{\psi} \circ \hat{f}$. Then $\widehat{\psi \circ f} = D(\psi \circ f)(x) \cdot \hat{\varphi}$.

The blowups of X_∞ and Y_∞ at x and $f(x)$ are (up to rescalings) limits of admissible inverse systems and φ and ψ blow up to height functions on these limits by [9], Theorem 9.15 (we need to exclude the case where x or $f(x)$ has height 0 or 1 if its corresponding system is unsigned). For a.e. $x \in X$, the blowup of U at x agrees with the blowup of X_∞ at x and the blowup of V at $f(x)$ agrees with the blowup of Y_∞ at $f(x)$.

Thus, $\hat{f}$ is a Lipschitz function between limits of admissible inverse systems that preserves height up to a constant scale (with norm equal to $|D(\psi \circ f)(x)| = |D\psi(f(x))Df(x)| = |Df(x)|$). By Lemma 2.10, it follows that $\hat{f}$ is $|Df(x)|$-Lipschitz.

Assuming also that $f(x)$ is a point of differentiability of f^{-1}, the preceding argument shows

that the inverse map of $\hat{f}$ is $|Df^{-1}(f(x))|$-Lipschitz. But $|Df^{-1}(f(x))| = |Df(x)|^{-1}$. Hence $\hat{f}$ is a $|Df(x)|$-homothety. This in turn implies that $\mathrm{Lip}_x f = |Df(x)|$. □

Theorem 3.3. *Let $f : U \subseteq X_\infty \to V \subseteq Y_\infty$ be a bi-Lipschitz homeomorphism between measurable subsets of Ahlfors Q-regular limit spaces of admissible inverse systems. Then, for any $u : V \to \mathbb{R}$ Lipschitz,*

$$\int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q = \int_V |Du|^Q \, d\mathcal{H}^Q.$$

Proof. By Lemma 3.1, Lemma 3.2, and (2.1)

$$\begin{aligned}
\frac{d\mathcal{H}^Q}{df_*\mathcal{H}^Q}(y) &\geq \frac{1}{(\mathrm{Lip}_y f^{-1})^Q} \\
&= 1/|Df^{-1}(y)|^Q \\
&= |Df(f^{-1}(y))|^Q
\end{aligned}$$

for a.e. $y \in V$. Then

$$\begin{aligned}
\int_V |Du|^Q \, d\mathcal{H}^Q &= \int_V |Du|^Q \frac{d\mathcal{H}^Q}{df_*\mathcal{H}^Q} \, df_*\mathcal{H}^Q \\
&\geq \int_V |Du(y)|^Q |Df(f^{-1}(y))|^Q \, df_*\mathcal{H}^Q(y) \\
&= \int_U |(Du)(f(x))|^Q |Df(f^{-1} \circ f(x))|^Q \, d\mathcal{H}^Q(x) \\
&\qquad \text{(applying the change of variables formula for pushforward measures)} \\
&= \int_U |(Du)(f(x))|^Q |Df(x)|^Q \, d\mathcal{H}^Q(x) \\
&= \int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q.
\end{aligned}$$

But we can repeat this argument for $f^{-1} : V \to U$ and $u \circ f : U \to \mathbb{R}$ to obtain

$$\int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q \geq \int_V |D(u \circ f \circ f^{-1})|^Q \, d\mathcal{H}^Q = \int_V |Du|^Q \, d\mathcal{H}^Q.$$

The theorem follows. □

3.2 Quasiconformal homeomorphisms

Suppose X, Y are complete Ahlfors Q-regular PI spaces that admit $(1, Q)$-Poincaré inequalities with $Q > 1$ (equivalently, complete Ahlfors Q-regular Loewner spaces). Also, suppose we are given open subsets $U \subseteq X$, $V \subseteq Y$ and a quasiconformal homeomorphism $f : U \to V$.

As in [7], Remark 4.45, which references the work in [11], Section 7, we can find a sequence of disjoint measurable sets $E_i \subseteq U$ such that $\mathcal{H}^Q(U \setminus \bigcup E_i) = 0$ and that the restriction of f to each E_i is a bi-Lipschitz embedding (note that we are applying Keith–Zhong self-improvement, since a strengthened Poincaré inequality is required than the one a priori given on X, Y, though we do not need Keith–Zhong self-improvement for admissible inverse systems). This allows us to define an induced map $f^* = f_x^* : T^*_{f(x)}Y \to T^*_x X$ for a.e. $x \in X$ in the following way: for a.e. $x \in X$, we have that x is contained in some E_i, and the restricton $f \mid E_i$ is a bi-Lipschitz homeomorphism $E_i \to f(E_i)$ that induces a mapping $T^*_{f(x)} f(E_i) \to T^*_x E_i$ (using that measurable subsets of a differentiability space inherit the differentiable structure in a natural way) which can be identified with a mapping $T^*_{f(x)}Y \to T^*_x X$; we define f_x^* to be this mapping obtained from $f \mid E_i$. If a decomposition other than $\{E_i\}$ were chosen, the induced mapping f_x^* would be the same for a.e. $x \in X$. We assume that this decomposition $\{E_i\}$ is fixed.

Assume X, Y have 1-dimensional cotangent bundle. Define $|Df(x)|$ to be the operator norm of f_x^*. By Lemma 3.2, for a.e. $x \in E_i$, we have $|Df(x)| = \mathrm{Lip}_x(f \mid E_i)$. If $g : V \to V'$ is a quasiconformal homeomorphism onto an open subset of a space satisfying the same hypotheses as

X, Y, then we have

$$|D(g \circ f)(x)| = |Dg(f(x))| \cdot |Df(x)|$$

for a.e. $x \in X$, and so

$$|Df^{-1}(y)| = 1/|Df(f^{-1}(y))| \tag{3.1}$$

for a.e. $y \in Y$.

If $u : V \to \mathbb{R}$ is Lipschitz, we can define

$$D(u \circ f)(x) = f_x^*(Du(f(x)))$$

for a.e. $x \in X$. This definition agrees with the original definition of $D(u \circ f)(x)$ if f were bi-Lipschitz (up to a set of measure zero). We can now state an analogue of Theorem 3.3 for quasiconformal homeomorphisms.

Theorem 3.4. *Let X_∞, Y_∞ be Ahlfors Q-regular limit spaces of admissible inverse systems with $Q > 1$ and $f : U \subseteq X_\infty \to V \subseteq Y_\infty$ a quasiconformal homeomorphism between open subsets.*

For any $u : V \to \mathbb{R}$ Lipschitz, we have

$$\int_U |D(u \circ f)|^Q \, d\mathcal{H}^Q = \int_V |Du|^Q \, d\mathcal{H}^Q.$$

Proof. Using the decomposition $\{E_i\}$ described in the beginning of this section, for each i,

$$\int_{E_i} |D(u \circ f)|^Q \, d\mathcal{H}^Q = \int_{f(E_i)} |Du|^Q \, d\mathcal{H}^Q$$

by Theorem 3.3, so the theorem follows by summing up over all i. □

We also show that $\operatorname{Lip}_x f = |Df(x)| = \operatorname{Lip}_x(f \mid E_i)$ and $|D(u \circ f)(x)| = \operatorname{Lip}_x u \circ f$ for a.e. $x \in E_i$ in the following lemma and theorem.

Lemma 3.5. *With f as in Theorem 3.4 and $\{E_i\}$ as described in the beginning of this section, for each i and a.e. $x \in E_i$,*

$$\operatorname{Lip}_x f = |Df(x)| = \operatorname{Lip}_x(f \mid E_i)$$

Proof. Recall that the second equality is by By Lemma 3.2.

By [11], Section 7, for a.e. $x \in E_i$, we have that $\operatorname{Lip}_x f$ and $\operatorname{Lip}_{f(x)} f^{-1}$ are finite. This is enough to ensure that, given a sequence of scale factors converging the zero, the blowup $\hat{f} : (\hat{X}, \hat{x}) \to (\hat{Y}, \hat{f(x)})$ of f at x is defined and is a quasisymmetric homeomorphism, as f is quasisymmetric in a neighborhood of x. Moreover, $\operatorname{Lip}_x f = \operatorname{Lip}_{\hat{x}} \hat{f}$. If we further assume that x is a point of Lebesgue density 1 of E_i, then $\operatorname{Lip}_{\hat{x}} \hat{f} = \operatorname{Lip}_x(f \mid E_i)$. The lemma follows. □

Theorem 3.6. *Let f be as in 3.4 and $\{E_i\}$ be as described in the beginning of this section. Suppose that $u : V \to \mathbb{R}$ is Lipschitz. For each i and a.e. $x \in E_i$, we have*

$$|D(u \circ f)(x)| = \operatorname{Lip}_x u \circ f.$$

In particular, by Theorem 3.4,

$$\int_U (\operatorname{Lip}_\bullet u \circ f)^Q \, d\mathcal{H}^Q = \int_V (\operatorname{Lip}_\bullet u)^Q \, d\mathcal{H}^Q.$$

Proof. By our definition of $D(u \circ f)(x)$ and $|Df(x)|$, for a.e. $x \in E_i$,

$$\begin{aligned} |D(u \circ f)(x)| &= |f_x^*(Du(f(x)))| \\ &= |Du(f(x))| \cdot |Df(x)| \\ &= \operatorname{Lip}_{f(x)} u \cdot \operatorname{Lip}_x f \\ &\geq \operatorname{Lip}_x u \circ f, \end{aligned}$$

where $|Du(f(x))| = \operatorname{Lip}_{f(x)} u$ by definition and $|Df(x)| = \operatorname{Lip}_x f$ by Lemma 3.5. But also,

$$\begin{aligned} |D(u \circ f)(x)| &= |f_x^*(Du(f(x)))| \\ &= |D((u \mid f(E_i)) \circ (f \mid E_i))(x)| \\ &= \operatorname{Lip}_x(u \mid f(E_i)) \circ (f \mid E_i) \\ &\leq \operatorname{Lip}_x u \circ f, \end{aligned}$$

where the "=" are by definition. The lemma follows. □

4 Preservation of Newtonian Q-energy and Q-harmonic functions

In this section we derive some consequences of our discussion in Section 3.2. We fix X_∞, Y_∞ Ahlfors Q-regular limit spaces of admissible inverse systems and a quasiconformal homeomorphism $f : U \subseteq X_\infty \to V \subseteq Y_\infty$ between bounded open subsets.

Our first goal is to prove an analogue of Theorem 3.4 for Newtonian Q-energy. We would like to know if $\|\rho_{u\circ f}\|_{L^Q(U)} = \|\rho_u\|_{L^Q(V)}$ holds for suitable u. As we prove shortly, this is indeed the case if u is in $N^{1,Q}(V)$ and bounded. We begin with a lemma.

Lemma 4.1. *Let $u \in N^{1,Q}(V)$. If u is Lipschitz and bounded, then $u \circ f \in N^{1,Q}(U)$ and*

$$\|\rho_{u\circ f}\|_{L^Q(U)} \leq \|\rho_u\|_{L^Q(V)}.$$

Proof. By [11], Theorem 8.1, f is absolutely continuous on Q-a.e. curve (note that the proof of this theorem implies a local result, so it is okay that f is not globally defined). Since u is locally Lipschitz, $u\circ f$ is absolutely continuous on Q-a.e. curve. With this in place, the proof of Proposition 1.11 in [7] can be applied to show that $\operatorname{Lip}_\bullet u \circ f$ is a weak upper gradient of $u \circ f$.

By Theorem 3.6,

$$\|\operatorname{Lip}_\bullet(u \circ f)\|_{L^Q(U)} = \|\operatorname{Lip}_\bullet u\|_{L^Q(V)}.$$

Observe that $\rho_u = \operatorname{Lip}_\bullet u$, because u is locally Lipschitz. Since $u \circ f$ is bounded, it follows that $u \circ f \in N^{1,Q}(U)$ and

$$\begin{aligned}\|\rho_{u\circ f}\|_{L^Q(U)} &\leq \|\operatorname{Lip}_\bullet(u \circ f)\|_{L^Q(U)} \\ &= \|\operatorname{Lip}_\bullet u\|_{L^Q(V)} \\ &= \|\rho_u\|_{L^Q(V)}.\end{aligned}$$

□

Theorem 4.2. *If $u \in N^{1,Q}(V)$ is bounded, then $u \circ f \in N^{1,Q}(U)$ and*

$$\|\rho_{u\circ f}\|_{L^Q(U)} = \|\rho_u\|_{L^Q(V)}.$$

Proof. If suffices to show that

$$\|\rho_{u\circ f}\|_{L^Q(U)} \leq \|\rho_u\|_{L^Q(V)},$$

because then we can repeat the argument with $u \circ f$ and f^{-1} to obtain

$$\|\rho_u\|_{L^Q(V)} = \|\rho_{u\circ f\circ f^{-1}}\|_{L^Q(V)} \leq \|\rho_{u\circ f}\|_{L^Q(U)}.$$

If $V = Y_\infty$, Theorem 2.12 in [22] states that there is a sequence of Lipschitz functions converging to u in $N^{1,Q}(V)$. We observe that in the proof of this theorem, if u is bounded to begin with, then the functions in this sequence can be chosen to be uniformly bounded. Theorem 5.47 in [1] generalizes this to arbitrary V open, but in this case the approximating sequence is locally Lipschitz; however, we still observe that, if u is bounded, the approximating sequence can be chosen to be uniformly bounded. Thus, we can construct a sequence $u_i \in N^{1,Q}(V)$ of locally Lipschitz and uniformly bounded functions converging to u in $N^{1,Q}(V)$. Furthermore, we may also that the convergence of this sequence occurs pointwise a.e.

First, by the bounded convergence theorem, $u_i \circ f$ converges to $u \circ f$ in $L^Q(U)$, since u_i converges to u a.e. Second, by Lemma 4.1, $\rho_{u_i\circ f}$ is a bounded sequence in $L^Q(U)$, and hence

converges weakly to some function g in $L^Q(U)$ that satisfies

$$\begin{aligned}\|g\|_{L^Q(U)} &\leq \liminf \|\rho_{u_i \circ f}\|_{L^Q(U)} \\ &\leq \liminf \|\rho_{u_i}\|_{L^Q(V)} \\ &= \|\rho_u\|_{L^Q(V)},\end{aligned}$$

where we have applied Lemma 4.1 in the second line and that $\|\rho_{u_i}\|_{L^Q(V)}$ converges to $\|\rho_u\|_{L^Q(V)}$ in the third line (the latter of which in turn is because $\|u_i\|_{N^{1,Q}(V)} - \|u_i\|_{L^Q(V)}$ converges to $\|u\|_{N^{1,Q}(V)} - \|u\|_{L^Q(V)}$). By [22], Lemma 3.6, these two observations together imply that g is a weak upper gradient of $u \circ f$. Thus, $u \circ f \in N^{1,Q}(U)$ and

$$\|\rho_{u\circ f}\|_{L^Q(U)} \leq \|g\|_{L^Q(U)} \leq \|\rho_u\|_{L^Q(V)}.$$

□

The preservation of Q-harmonic functions is a quick corollary.

Theorem 4.3. *If $u \in N^{1,Q}(V)$ is bounded and Q-harmonic, then $u \circ f \in N^{1,Q}(U)$ is Q-harmonic.*

Proof. If suffices to show, for bounded $\varphi \in N^{1,Q}(U)$ with compact support in U, that $\|\rho_{u\circ f}\|_{L^Q(U)} \leq \|\rho_{u\circ f+\varphi}\|_{L^Q(U)}$. For such φ,

$$\begin{aligned}\|\rho_{u\circ f}\|_{L^Q(U)} &= \|\rho_u\|_{L^Q(V)} \\ &\leq \|\rho_{u+\varphi\circ f^{-1}}\|_{L^Q(V)} \\ &= \|\rho_{u\circ f+\varphi}\|_{L^Q(U)}.\end{aligned}$$

The "="s are by Theorem 4.2; the "≤" is due to u being Q-harmonic in V and (by Theorem 4.2) $\varphi \circ f^{-1}$ being a bounded function in $N^{1,Q}(V)$ with compact support in V. □

5 Rigidity of quasiconformal homeomorphisms of limit spaces of a class of admissible inverse systems

5.1 A class $\mathcal{F}$ of admissible inverse systems

We define a certain class $\mathcal{F}$ of admissible inverse systems for which we apply the results of the previous sections to obtain rigidity results. The unsigned inverse system $(X_\bullet, d_\bullet, \mu_\bullet, \pi_\bullet)$ is said to be in the class $\mathcal{F}$ if the conditions below hold for some fixed parameters $m, k \geq 2$. (Recall that X_i' is the graph obtained by subdividing each edge of X_i into m parts.)

(B1) (X_0, d_0, μ_0) is isomorphic to the unit interval

(B2) Each edge of X_i has d_i-length m^{-i}. The restriction of μ_i to each edge of X_i is rescaled length measure, with the same scaling on each edge, normalized so that $\mu_i(X_i) = 1$

(B3) The graph X_{i+1} is a quotient graph of k isomorphic copies $X_{i,1}', \ldots, X_{i,k}'$ of X_i' with isomorphisms $\sigma_j : X_{i,j}' \to X_i'$. We identify $x \in X_{i,j_1}', y \in X_{i,j_2}'$ if $\sigma_{j_1}(x) = \sigma_{j_2}(y)$ and $\sigma_{j_1}(x)$ is a vertex of X_i' that lies in the interior of an edge in X_i; all other points are identified only with themselves. If $\tau_{i+1} : \bigsqcup_j X_{i,j}' \to X_{i+1}$ denotes the quotient map, then $\pi_i \circ (\tau_{i+1} \mid X_{i,j}')$ is equal to the isomorphism σ_j

The idea of these constructions goes back to the Laakso spaces introduced in [16], with the case $k = 2$ obtained in this paper as quotients of products of Cantor sets and intervals. It was known that Laakso spaces could be obtained as limits of sequences of graphs. J. Cheeger and B. Kleiner generalized these and introduced axioms for inverse systems of graphs in [8] whose limit spaces provide a plethora of examples of PI spaces.

It is a routine verification that inverse systems in the class $\mathcal{F}$ satisfy the axioms for admissible inverse systems.

Lemma 5.1. *With τ_{i+1} as above, we have that $\tau_{i+1} \mid X_{i,j}'$ is an isometric embedding.*

Proof. Let U denote the image of $\tau_{i+1} \mid X'_{i,j}$, and consider a geodesic $\gamma : [0, L] \to X_{i+1}$ joining two points of U. Then $(\tau_{i+1} \mid X'_{i,j}) \circ \sigma_j^{-1} \circ \pi_i \circ \gamma$ is a path in U of the same length as γ joining $\gamma(0)$ to $\gamma(L)$. Since $\tau_{i+1} \mid X'_{i,j}$ is 1-Lipschitz, the lemma follows. □

Proposition 5.2. *For each i, there is an isometric section of the projection map $\Pi_i : X_\infty \to X_i$. The union of the images of these sections is dense in X_∞.*

Proof. Starting with X_i, we can lift this entire graph in an isometric way as a section of the projection $\pi_i : X_{i+1} \to X_i$ so that the image of this section is the image of some $\tau_{i+1} \mid X'_{i,j}$; this is an isometric lift by Lemma 5.1. Then we lift X_{i+1} in the same way, then X_{i+2}, and so on. Taking the inverse limit, this gives us an isometry of X_i into X_∞ as a section of Π_i. By axiom (3) for admissible inverse systems, the union of the images of all of these sections is dense in X_∞. □

Fix an unsigned inverse system $(X_\bullet, \pi_\bullet)$ in the class $\mathcal{F}$ with parameters $m, k \geq 2$. To apply the results of Section 4, we first need to verify that the measure μ_∞ corresponds to $\mathcal{H}^Q$ (rescaled) for some $Q > 1$.

Fix $Q > 1$ so that

$$mk \left(\frac{1}{m}\right)^Q = 1. \tag{5.1}$$

Explicitly, $Q = 1 + \log(k)/\log(m)$.

Recall the projections Π_j^{i+j} and $\Pi_j : X_\infty \to X_j$ from 2.4 and 2.5. Note that X_j has $(mk)^j$ edges and, for each edge $e_j \subseteq X_j$,

$$\begin{aligned} \mu_\infty(\Pi_j^{-1}(e_j)) &= \mu_j(e_j) \\ &= \left(\frac{1}{mk}\right)^j \\ &= \left(\frac{1}{m^j}\right)^Q \end{aligned}$$

by the previous paragraph. Also, as a consequence of axiom (3) for admissible inverse systems,

there is a constant $C_1 \geq 1$ such that

$$m^{-j} \leq \operatorname{diam}(\Pi_j^{-1}(e_j)) \leq C_1 m^{-j}.$$

Lemma 5.3. *The measure μ_∞ is Ahlfors Q-regular.*

Proof. Let $x \in X_\infty$ and $0 < r < 1$. Choose j so that

$$C_1 m^{-j} \leq r/2 \leq C_1 m^{-(j-1)},$$

with C_1 as in the paragraph preceding the statement of this lemma. If $e_j \subseteq X_j$ denotes an edge containing $\Pi_j(x)$, it follows that $B(x, r)$ contains $\Pi_j^{-1}(e_j)$. Hence

$$\begin{aligned} \mu_\infty(B(x,r)) &\geq \mu_\infty(\Pi_j^{-1}(e_j)) \\ &= \mu_j(e_j) \\ &= m^{-jQ} \\ &\geq C_2^{-1} r^Q. \end{aligned}$$

Next, choose j so that

$$m^{-(j+1)} \leq r < m^{-j}.$$

Denote $x_j = \Pi_j(x)$. Since Π_j is 1-Lipschitz, we have $\Pi_j(B(x, r)) \subseteq B(x_j, r)$. If $e_j \subseteq X_j$ denotes an edge containing x_j, since $r < m^{-j}$, we see that $B(x_j, r)$ is contained in the union of e_j and its adjacent edges in X_j. Since each vertex in X_j has degree $\leq 2k$, we then have that $B(x_j, r)$ is contained in a set S that is the union of at most $2 \cdot 2k$ edges in X_j, and so $\mu_j(S) \leq 2 \cdot 2k \cdot m^{-jQ}$.

Finally,

$$\begin{aligned}\mu_\infty(B(x,r)) &\le \mu_\infty(\Pi_j^{-1}(B(x_j,r)))\\ &= \mu_j(B(x_j,r))\\ &\le \mu_j(S)\\ &\le 2\cdot 2k\cdot m^{-jQ}\\ &\le C_2 r^Q.\end{aligned}$$

□

Fix i and consider an edge $e_i \in X_i$. Let $Y_0 = e_i$ and $Y_{j+1} = \pi_{i+j}^{-1}(Y_j)$. Restricting π_{i+j} to Y_{j+1} gives us an inverse system $Y_\bullet$ that is isomorphic as a system of graphs to $X_\bullet$, but with edges rescaled by m^{-i}.

Using the length metric on each Y_k and taking the inverse limit, we obtain the metric space Y_∞ which is m^{-i}-homothetic to X_∞. The graph embeddings $F_j : Y_j \to X_{i+j}$ induce a mapping $F : Y_\infty \to X_\infty$ that is easily shown to be a local isometry on $Y_\infty \setminus \{\text{points of height 0 or 1}\}$. We show F is actually an isometry.

Lemma 5.4. *The mapping $F : Y_\infty \to X_\infty$ in the previous paragraph is an isometry.*

Proof. We need only show that each F_j is an isometry. It suffices to show that any two distinct points of $F_j(Y_j)$ can be joined by a geodesic in $F_j(Y_j)$, but this is due to two observations: Let v_1 and v_2 denote the two vertices of e_i.

- An inductive argument shows that two points of $(\Pi_i^j)^{-1}(v_\ell)$ can be joined by a geodesic contained in $F_j(Y_j)$.
- Points of $(\Pi_i^j)^{-1}(v_1)$ can be joined to points of $(\Pi_i^j)^{-1}(v_2)$ with monotone paths contained in $F_j(Y_j)$, and these paths must be geodesics.

If there is a geodesic γ joining two points of $F_j(Y_j)$, on any interval on which this geodesic leaves then re-enters $F_j(Y_j)$, the restriction of γ to this interval joins two points of $(\Pi_i^j)^{-1}(v_1) \cup (\Pi_i^j)^{-1}(v_2)$. But then we may modify γ by the above observations to a geodesic that does not leave $F_j(Y_j)$ on this interval. □

Proposition 5.5. *The measure μ_∞ coincides with $\mathcal{H}^Q$ (rescaled).*

Proof. By Lemma 5.3, there is a constant $C_3 \geq 1$ so that $C_3^{-1}\mu_\infty \leq \mathcal{H}^Q \leq C_3\mu_\infty$, so in particular the $\mathcal{H}^Q$-measure of an open set in X_∞ is positive and finite. It suffices to show there exists a positive constant C_4 which satisfies, for each i and edge $e_i \subseteq X_i$, that $\mathcal{H}^Q(\Pi_i^{-1}(e_i)) = C_4 \cdot \mu_\infty(\Pi_i^{-1}(e_i)) = C_4 m^{-iQ}$ (since any open set in X_∞ can be written as a countable union of sets of the form $\Pi_i^{-1}(e_i)$ with possibly varying i). We show that $C_4 = \mathcal{H}^Q(X_\infty)$ suffices.

Fix i and consider an edge $e_i \in X_i$. By the previous lemma, $\Pi_i^{-1}(e_i)$ is isometric to X_∞ scaled by m^{-i}, and so

$$\begin{aligned}\mathcal{H}^Q(\Pi_i^{-1}(e_i)) &= m^{-iQ}\mathcal{H}^Q(X_\infty) \\ &= C_4 m^{-iQ}.\end{aligned}$$

□

We next verify a property of the isometry group of X_∞. We use the notation $X'_{i,j}$, σ_j, and τ_i from the beginning of this section.

Proposition 5.6. *Fix an edge $e_i \subseteq X_i$ and let $U = \Pi_i^{-1}(e_i)$. If $x, y \in U$ are such that $\Pi_i(x) = \Pi_i(y)$, then there exists an isometry of X_∞, equal to the identity outside the $m^{-(i+1)}$-neighborhood of U, which maps x into y.*

Proof. Recall that $\Pi_i^{i+j} = \pi_i\pi_{i+1}\cdots\pi_{i+j-1} : X_{i+j} \to X_i$ and let $U_{i+j} = (\Pi_i^{i+j})^{-1}(e_i)$. It suffices to show that there exists a compatible sequence of graph isomorphisms $f_{i+j} : X_{i+j} \to X_{i+j}$ mapping

$\Pi_{i+j}(x)$ into $\Pi_{i+j}(y)$ that is the identity outside the $m^{-(i+1)}$-neighborhood of U_{i+j} for each $j \geq 1$. By "compatible," we mean that if f_{i+j+1} maps x' into y', then f_{i+j} maps $\pi_{i+j}(x')$ into $\pi_{i+j}(y')$. We proceed with this proof by induction on j.

For the base case $j = 1$, let $x_{i+1} = \Pi_{i+1}(x)$ and $y_{i+1} = \Pi_{i+1}(y)$. If $x_{i+1} \neq y_{i+1}$, there are three cases to consider.

First, suppose x_{i+1} lies in an edge of X_{i+1} that is mapped into the interior of e_i by π_i. Then x_{i+1} and y_{i+1} lie in a cycle subgraph of two edges in U_{i+1} which is collapsed to a single edge of X_i' by π_i; it is clear an isomorphism of X_{i+1} of the desired form exists.

Second, suppose x_{i+1} lies in an edge of X_{i+1} whose image in X_i under π_i intersects a vertex of e_i that is of height 0 or 1. The argument is similar to the first case, except that x_{i+1} and y_{i+1} lie in a path subgraph of two edges in U_{i+1} which is collapsed to a single edge of X_i' by π_i.

Third, suppose x_{i+1} lies in an edge of X_{i+1} whose image in X_i under π_i intersects a vertex of e_i that is not of height 0 or 1. Then x_{i+1} and y_{i+1} lie in a subgraph G of X_{i+1} obtained by gluing two isomorphic star graphs of $2k$ edges along corresponding degree 0 vertices; the quotient map π_i collapses this subgraph to a star subgraph of X_i' of $2k$ edges. The isomorphism of G swapping the two isomorphic star graphs maps x_{i+1} into y_{i+1} and extends to an isomorphism of X_{i+1} that is the identity outside of G. Elements of G not in U_{i+1} are of distance at most $m^{-(i+1)}$ from U_{i+1}, so we have found an isomorphism of the desired form.

For the induction step, let $x_{i+j+1} = \Pi_{i+j+1}(x)$ and $y_{i+j+1} = \Pi_{i+j+1}(y)$. There is an obvious isomorphism $\bar{f}_{i+j+1} : X_{i+j+1} \to X_{i+j+1}$ compatible with f_{i+j}: consider the isomorphisms $\sigma_k : X'_{i+j,k} \to X'_{i+j}$. We can apply $\sigma_k^{-1} \circ f_{i+j} \circ \sigma_k$ to $X'_{i+j,k}$ for each k to define an isomorphism $\bigsqcup X'_{i+j,k} \to \bigsqcup X'_{i+j,k}$ which preserves the fibers of the projection map $\bigsqcup X'_{i+j,k} \to X_{i+j+1}$, and hence descends to $\bar{f}_{i+j+1} : X_{i+j+1} \to X_{i+j+1}$ compatible with f_{i+j}. Notice that $\bar{f}_{i+j+1}$ maps U_{i+j+1} into U_{i+j+1} and is the identity outside of the $m^{-(i+1)}$-neighborhood of U_{i+j+1}.

Now, $\bar{f}_{i+j+1}$ maps x_{i+j+1} into y'_{i+j+1} with $\pi_{i+j}(y'_{i+j+1}) = \pi_{i+j}(y_{i+j+1})$. We can adapt the reasoning in the base case and compose $\bar{f}_{i+j+1}$ with an isometry that maps y'_{i+j+1} into y_{i+j+1} and is

the identity outside the $m^{-(i+j)}$-neighborhood of U_{i+j+1} to obtain f_{i+j+1}. □

Corollary 5.7. *With e_i and U as in Proposition 5.6, suppose that e_i has a vertex of height* 0 *or* 1*. Then the isometry in the conclusion of this proposition is the identity on points of height* 0 *or* 1 *not in U.*

Proof. Let $U_j = (\Pi_i^j)^{-1}(e_i)$. Then the corollary follows from the fact that the points of height 0 or 1 that are not in U_j are of distance $\geq m^{-i}$ from U_j. □

We also need to study the structure of a specific type of mapping between open subsets of X_∞. By a *vertex level* in X_∞ we mean a point x in X_∞ such that $\Pi_i(x)$ is a vertex of X_i for some i. Equivalently, $\varphi(x)$ is an m-adic rational number. Note that these are called "wormholes" in [16]. An edge preimage subspace of level i, shortened to *edge preimage*, is a preimage in X_∞ of an edge of the graph X_i under the projection Π_i.

Proposition 5.8. *Fix an edge $e_i \subseteq X_i$ and let $U = \Pi_i^{-1}(\text{interior}(e_i))$. Let $f : U \to X_\infty$ be a continuous open mapping that is affine on height, that is, there exists $\lambda \in \mathbb{R} \setminus \{0\}$ and $b \in \mathbb{R}$ such that $\varphi(f(x)) = \lambda\varphi(x) + b$. Then:*

(1) f maps vertex levels into vertex levels and non-vertex levels into non-vertex levels.

(2) $|\lambda|$ is equal to $m^{-\ell}$ for some $\ell \in \mathbb{Z}$.

(3) If $m \geq 3$, then f establishes an $m^{-\ell}$-homothety between U and an edge preimage of level $i + \ell$

(4) If $m = 2$, either the conclusion of (3) holds or, for each edge e_{i+1} in $\pi_i^{-1}(e_i) \subseteq X_{i+1}$, f establishes an $m^{-\ell}$ homothety between the preimage subspace of e_{i+1} and an edge preimage of level $i + 1 + \ell$

Proof. (1) By Remark 5.1, Lemma 5.4, and considering f^{-1}, we see that f is a local $|\lambda|$-homothety that preserves differences in height up to the constant λ. Note f cannot map into the points of height

0 or 1 as any neighborhood of a point in U contains points of higher and lower height. Then, by Lemma 5.11, we see that f maps vertex levels into vertex levels. The same holds for f^{-1}, so f maps non-vertex levels into non-vertex levels.

(2) Since vertex levels exist only at m-adic rational heights, we see that $\lambda = r/m^\ell$ with $r, \ell \in \mathbb{Z}$ and r coprime to m. Observe that f^{-1} satisfies similar hypotheses as f when restricted to the preimage $\Pi_j^{-1}(\text{interior}(e_j))$ of the interior of a suitable edge e_j in X_j; since f^{-1} preserves differences in height up to the constant m^ℓ/r, it follows that $r \in \{\pm 1\}$.

(3) Suppose that $m \geq 3$. Observe that $\varphi(U)$ is an interval (a, b). We have a well-defined linear map $F : (a, b) \to [0, 1]$ given by $F(c) = \varphi(f(x))$ for any $x \in U$ such that $\varphi(x) = c$, which extends to a linear map $F : [a, b] \to [0, 1]$. Note that

$$(b - a) = m^{-i}, \qquad |F(b) - F(a)| = m^{-(i+\ell)}. \tag{5.2}$$

Note that a is an integer multiple of m^{-i}. Also let $[A, B] = F([a, b])$.

We claim that $F(a)$ is an integer multiple of $m^{-(i+\ell)}$. If not, (A, B) contains an integer multiple of $m^{-(i+\ell)}$. Hence $\Pi_{i+\ell}(U)$ contains a vertex $v_{i+\ell}$ of $X_{i+\ell}$ (not of height 0 or 1). We have two observations:

- Since $\Pi_{i+\ell}$ is an open map, $\Pi_{i+\ell}(U)$ contains a neighborhood of $v_{i+\ell}$ in $X_{i+\ell}$.
- Given any two edges in the star of $v_{i+\ell}$ of distinct heights except the height of this vertex, there is a monotone geodesic $\gamma : (A, B) \to \Pi_{i+\ell}(U)$ intersecting these two edges.

If $F(a)$ is not an integer multiple of $m^{-(i+\ell)}/2$, we see that $\Pi_{i+\ell}(U)$ contains the midpoints of two distinct edges of $X_{i+\ell}$, in the star of a vertex, and these midpoints have the same height. These midpoints lie at a distance $m^{-(i+\ell)}$ apart. On the other hand, f is an $m^{-\ell}$-Lipschitz map, and by Lemmas 5.4 and 5.12 we have that points of the same height in U are mapped to points of distance $< m^{-(i+\ell)}$ apart, and so points in $\Pi_{i+\ell}(U)$ of the same height are of distance $< m^{-(i+\ell)}$ apart.

On the other hand, if $F(a)$ is equal to $p \cdot m^{-(i+\ell)}/2$ with p coprime to m and 2, then a point of height $a + m^{-i}$ (which is an integer multiple of m^{-i}) in U is mapped to a point of height $p \cdot m^{-(i+\ell)}/2 \pm m^{-(i+\ell)}$ by f, so f maps a vertex level to a non-vertex level.

Thus, $F(a)$ and $F(b)$ are integer multiples of $m^{-(i+\ell)}$. A connectedness argument shows that $\Pi_{i+\ell}(U)$ is the full interior of an edge $\tilde{e}_{i+\ell} \subseteq X_{i+\ell}$. Noting that U has the same Hausdorff measure as $V = \Pi_{i+\ell}^{-1}(\text{interior}(\tilde{e}_{i+\ell}))$, it follows that U is a dense subset of V, from which it follows that $U = V$. Then $f : U \to V$ is an $m^{-\ell}$-Lipschitz homeomorphism and f^{-1} is an m^{ℓ}-Lipschitz homeomorphism, so that $f : U \to V$ is an $m^{-\ell}$-homothety.

(4) If $m = 2$, we can attempt to repeat the argument in (3), but the conclusion of (3) will not hold when $F(a)$ is not an integer multiple of $m^{-(i+\ell)}$ but is an integer multiple of $m^{-(i+\ell)}/2$. Assuming this, given an edge $e_{i+1} \subseteq \pi_i^{-1}(e_i)$, a similar conclusion as (3) holds assuming we replace f with its restriction to $\Pi_{i+1}^{-1}(\text{interior}(e_{i+1}))$. □

Corollary 5.9. *Assume $m \geq 3$. Let i, e_i, U, f, ℓ be as in Proposition 5.8 and $\tilde{e}_{i+\ell}$ the edge associated with $f(U)$. Define*

$$U_i = e_i, \quad U_{j+1} = \pi_j^{-1}(U_j), \quad V_{i+\ell} = \tilde{e}_{i+\ell}, \quad V_{j+1+\ell} = \pi_{j+\ell}^{-1}(V_{j+\ell}).$$

Then f is induced by a compatible system of graph isomorphisms $f_j : U_j \to V_{j+\ell}$ for $j \geq i$ such that

- *$f_{j+1}(x) = y$ if and only if $f_j(\pi_j(x)) = \pi_{j+\ell}(y)$ and*
- *$f(x) = y$ if and only if $f_j(\Pi_j(x)) = \Pi_{j+\ell}(y)$.*

Proof. This follows by applying Proposition 5.8 to each edge in U_j. The only subtle point we need to check is that, given an edge $e_j \subseteq U_j$, if f maps $\Pi_j^{-1}(\text{interior}(e_j))$ onto $\Pi_{j+\ell}^{-1}(\text{interior}(\tilde{e}_{j+\ell}))$ for some edge $\tilde{e}_{j+\ell} \subseteq V_{j+\ell}$, letting e_{j+1}^r (resp. $\tilde{e}_{j+\ell+1}^r$) denote the edges in $\pi_j^{-1}(e_j)$ (resp. $\pi_{j+\ell}^{-1}(\tilde{e}_{j+\ell})$),

we have that $\Pi_{j+\ell+1} \circ f(\bigcup_{r=1}^{mk} \Pi_{j+1}^{-1}(\text{interior}(e_{j+1}^r))) = \bigcup_{r=1}^{mk} \text{interior}(\tilde{e}_{j+\ell+1}^r)$. But this latter statement follows from the fact that $\Pi_{j+\ell+1} \circ f$ is an open map. □

For the case $m = 2$, similar reasoning shows:

Corollary 5.10. *Assume $m = 2$ and let f be as in Proposition 5.8. If the conclusion of Proposition 5.8(3) holds for f, then the conclusion of Corollary 5.9 holds. If the conclusion of Proposition 5.8(3) does not hold for f, then f is still similarly induced by a compatible system of graph isomorphisms $f_j : U_j \to V_{j+\ell}$ for $j \geq i+1$, where in this case $U_{i+1} = \pi_i^{-1}(e_i)$.*

Now we prove the lemmas used in the previous discussion: Consider a geodesic $\gamma : [-\varepsilon, \varepsilon] \to X_\infty$ with $\gamma(0) = x$ such that $\gamma \mid [-\varepsilon, 0]$ and $\gamma \mid [0, \varepsilon]$ are monotone but traversing in opposite directions. Call such a geodesic a "two-piece-monotone geodesic that switches direction at x."

Lemma 5.11. *If $x \in X_\infty$ is a vertex level not of height 0 or 1, then there exists a two-piece-monotone geodesic that switches direction at x. For all other $x \in X_\infty$, there does not exist a two-piece-monotone geodesic that switches direction at x.*

The idea for this lemma comes from the structure of geodesics in Laakso spaces ([16], Proposition 1.1).

Proof. Suppose that $x \in X_\infty$ is such that $v_i = \Pi_i(x)$ is a vertex in X_i not of height 0 and 1. A neighborhood of $\Pi_i(x)$ in X_i is isometric to $2k$ intervals joined at the point v_i, and so there exists a geodesic $\gamma_i : [-\varepsilon, \varepsilon] \to X_i$ with $\gamma_i(0) = v_i$ such that $\gamma_i \mid [-\varepsilon, 0]$ and $\gamma_i \mid [0, \varepsilon]$ are monotone but traversing in opposite directions. Using Proposition 5.2, we can lift γ_i to geodesics $\gamma_j : [-\varepsilon, \varepsilon] \to X_j$ such that $\Pi_i^j \circ \gamma_j = \gamma_i$. Then the inverse limit of the γ_j is a two-piece-monotone geodesic $\gamma : [\varepsilon, \varepsilon] \to X_\infty$ with $\gamma(0) = x$.

On the other hand, suppose that $x \in X_\infty$ is not as in the first sentence of the previous paragraph. Let $\gamma : [-\varepsilon, \varepsilon] \to X_\infty$ be a two-piece-monotone geodesic that switches direction at x. Let i be such that $v_i = \Pi_i(\gamma(-\varepsilon))$ and $w_i = \Pi_i(\gamma(\varepsilon))$ are distinct. Further assume that $\varepsilon \geq 2 \cdot m^{-i}$.

We have that $\gamma_i = \Pi_i \circ \gamma$ is a piecewise monotone path, traversing in different directions on $[-\varepsilon, 0]$ and $[0, \varepsilon]$. By the choice of x, there is some $\alpha \in (0, \varepsilon)$ so that $\gamma_i(-\alpha) = \gamma_i(\alpha) = v_i'$ is a vertex of X_i and $\varepsilon' = \varepsilon - \alpha \geq m^{-i}$. Thus, γ_i can be shortened to a path $\tilde{\gamma}_i : [-\varepsilon', \varepsilon']$ joining v_i to w_i with $\gamma(0) = v_i'$, monotone on $[-\varepsilon', 0]$ and $[0, \varepsilon']$, and traversing in different directions on each. Note $\varepsilon' \geq m^{-i} \geq 2 \cdot m^{-(i+1)}$, from which it follows by Lemma 5.13 that $\tilde{\gamma}_i$ can be lifted, for $j > i$, to a path in each X_j joining $\Pi_j(\gamma(-\varepsilon))$ to $\Pi_j(\gamma(\varepsilon))$. Taking the inverse limit of the $\tilde{\gamma}_j$, we obtain a path in X_∞ joining $\gamma(-\varepsilon)$ to $\gamma(\varepsilon)$ with smaller length than γ. Thus no such path γ exists. □

Lemma 5.12. *For any $c \in (0, 1)$, we have that $diam(\varphi^{-1}(\{c\})) < 1$.*

Proof. Suppose that $x, y \in X_\infty$ are such that $\varphi(x) = \varphi(y)$ and let $x_i = \Pi_i(x)$ and $y_i = \Pi_i(y)$. First assume $m > 2$. In X_0, there is a two-piece monotone path (it is not an injective path) $\gamma_0 : [-m^{-1}, m^{-1}] \to X_0$ joining x_0 to y_0 (actually $x_0 = y_0$), monotone on $[-m^{-1}, 0]$ and $[0, m^{-1}]$, and traversing in different directions on these two intervals. By Lemma 5.13, γ_0 lifts to a two-piece monotone path joining x_i to y_i of length $2m^{-1}$. Taking the inverse limit of the γ_i, we obtain a path γ joining x to y of length $2m^{-1} < 1$.

If $m = 2$ and $\varphi(x) \neq 1/2$, choose the two-piece monotone path $\gamma_0 : [-L, L] \to X_0$ joining x_0 to y_0 so that it passes through a point of height $1/2$ and satisfies $1/2 < 2L < 1$. For $i \geq 1$, we have that $2L > 2 \cdot m^{-(i+1)}$, and noting that γ_0 passes through the center vertex of X_0', we see that γ_0 lifts to a two-piece monotone path joining x_i to y_i of length $2L$ for all $i \geq 1$. Again, taking the inverse limit of the γ_i, we obtain a path γ joining x to y of length $2L < 1$.

The case $m = 2$ and $\varphi(x) = 1/2$ then follows from Lemma 5.4. □

Lemma 5.13. *Let $x_{i+1}, y_{i+1} \in X_{i+1}$ and $x_i = \pi_i(x_{i+1}), y_i = \pi_i(y_{i+1})$. Given a monotone path γ_i joining x_i to y_i, we can lift γ_i to a monotone path γ_{i+1} with $\pi_i \circ \gamma_{i+1} = \gamma_i$ and γ_{i+1} joining x_{i+1} to y_{i+1} if the image of γ_i contains a vertex of X_i' that is not a vertex of X_i. In particular, this occurs when $length(\gamma_i) \geq 2 \cdot m^{-(i+1)}$ or if $i = 0$ and $length(\gamma_i) \geq m^{-1}$.*

Proof. Let T denote the vertices of X_{i+1} that do not project to vertices of X_i under π_i. If $x_{i+1} \in T$ or $y_{i+1} \in T$, the lemma follows easily, so assume this is not the case. We know that we can lift γ_i to a monotone path $\tilde{\gamma}_{i+1}$ joining x_{i+1} to some $\tilde{y}_{i+1}$ with $\pi_i(\tilde{y}_{i+1}) = y_i$. Then x_{i+1} and y_{i+1} lie in different components of $X_{i+1} \setminus T$ and y_{i+1} and $\tilde{y}_{i+1}$ lie in the same component. There is a height-preserving isometry of X_{i+1} which maps y_{i+1} into $\tilde{y}_{i+1}$ and is the identity outside of the component of $X_{i+1} \setminus T$ containing y_{i+1}. Composing $\tilde{\gamma}_{i+1}$ with this isometry gives us the desired path γ_{i+1}. □

5.2 Q-harmonic functions on limit spaces of inverse systems in $\mathcal{F}$

Fix an unsigned inverse system $(X_\bullet, \pi_\bullet)$ in the class $\mathcal{F}$ with parameters $m, k \geq 2$ and define Q as in 5.1. Let $u : U \subseteq X_\infty \to \mathbb{R}$ be continuous and Q-harmonic in an open set U. If e_i denotes an edge of X_i that is contained in $\Pi_i(U)$, then if $i \geq 1$ is sufficiently large, the $m^{-(i+1)}$-neighborhood of $V_i = \Pi_i^{-1}(e_i)$ is compactly contained in U.

We first show that Q-harmonic functions on X_∞ descend to its graph approximations in the following sense:

Proposition 5.14. *Let u, i, and V_i be as above. If $x, y \in V_i$ are such that $\Pi_i(x) = \Pi_i(y)$, then $u(x) = u(y)$.*

Proof. By Proposition 5.6, there is an isometry $f : X_\infty \to X_\infty$ mapping x into y that is the identity outside the $m^{-(i+1)}$-neighborhood of V_i. Then note that $u \circ f$ is a continuous Q-harmonic function in U that agrees with u outside the $m^{-(i+1)}$-neighborhood of V_i. The proposition follows from the uniqueness part of Theorem 2.7. □

Actually, Q-harmonic functions on X_∞ descend to linear functions in the following sense:

Proposition 5.15. *Let u, i, e_i, and V_i be as above. Let $v_1, v_2 \in X_i$ denote the vertices of e_i with heights $a < b$. Let $L : [0,1] \to \mathbb{R}$ denote the linear function with $L(a) = u(x)$ and $L(b) = u(y)$ for any $x \in \Pi_i^{-1}(\{v_1\})$ and $y \in \Pi_i^{-1}(\{v_2\})$(by the previous proposition, L is well-defined). Let $\ell(x) = L \circ \varphi_\infty(x)$ for $x \in X_\infty$. Then $u(x) = \ell(x)$ for $x \in V_i$.*

Note that ℓ can possibly be a constant function.

Proof. This follows from Lemmas 2.9 and 2.11 as the function $\tilde{u}$ equal to ℓ in V_i and u outside V_i has Q-energy at most that of u and agrees with u outside a compact set in U. □

Remark 5.16. The discussion in this section holds for Q'-harmonic functions with $Q' \in (1, \infty)$.

5.3 Quasiconformal homeomorphisms

As before, fix an unsigned inverse system $(X_\bullet, \pi_\bullet)$ in the class $\mathcal{F}$ with parameters $m, k \geq 2$ and define Q as in 5.1. Suppose that $f : U \to V$ is a quasiconformal homeomorphism between open subsets of X_∞. Let $S = \varphi^{-1}(\{0\}) \cup \varphi^{-1}(\{1\})$.

Lemma 5.17. *f bijectively maps $U \cap S$ onto $V \cap S$.*

Proof. If $x \in U \setminus S$, then there exists a continuous Q-harmonic function defined in a neighborhood of x that is non-constant in any neighborhood of x contained in U. On the other hand, if $x \in U \cap S$, given a continuous Q-harmonic function u defined in a neighborhood of x, by Proposition 5.15, there is an edge $e_i \subseteq X_i$ such that $U_i = \Pi_i^{-1}(e_i)$ is a neighborhood of x compactly contained in U and such that $u(y)$ for $y \in U_i$ only depends on $\Pi_i(y)$; note the corresponding function on e_i is linear. Since $\Pi_i(x)$ is a vertex of X_i, it is easy to see that there is a variation of u that is constant in U_i and equal to u outside U_i. Hence u is constant in U_i.

If u (resp. v) is Q-harmonic in a neighborhood of x (resp. $f(x)$) contained in U (resp. V), then by Theorem 4.3, $u \circ f$ (resp. $v \circ f^{-1}$) is Q-harmonic in a neighborhood of $f(x)$ (resp. x) contained in V (resp. U). By the previous paragraph, the lemma follows. □

By Lemma 2.11, Theorem 4.3, and Lemma 5.17, we see that $\varphi \circ f$ is Q-harmonic in $U \setminus S$. Let $x \in U$. If $\varphi(x) \notin \{0, 1\}$, given an edge $e_i \subseteq X_i$ containing $\Pi_i(x)$ such that the $m^{-(i+1)}$-neighborhood of $V_i = \Pi_i^{-1}(e_i)$ is contained in U and such that neither vertex of e_i has height 0 or

1, then $\varphi \circ f(y)$ depends only on $\Pi_i(y)$ for $y \in V_i$ and the corresponding function on e_i is linear by Proposition 5.15.

On the other hand, if $\varphi(x) \in \{0, 1\}$, if i is sufficiently large, given the edge $e_i \subseteq X_i$ containing $\Pi_i(x)$ and defining $V_i = \Pi_i^{-1}(e_i)$ as before, we have that $\varphi \circ f$ is constant on $V_i \cap S$, by Lemma 5.17 (assume that only one of the two vertices of e_i has height 0 or 1), and the $m^{-(i+1)}$-neighborhood of V_i is contained in U. If $y, z \in V_i$ are such that $\Pi_i(y) = \Pi_i(z)$, then there is an isometry h of X_∞ mapping $y \to z$ that is the identity outside of the $m^{-(i+1)}$-neighborhood of V_i and is the identity outside of $V_i \cap S$, by Proposition 5.6 and Corollary 5.7. Then $\varphi \circ f$ and $\varphi \circ f \circ h$ are Q-harmonic in $U \setminus S$ and agree outside of the $m^{-(i+1)}$-neighborhood of V_i (which is contained in $U \setminus S$, but does not have compact closure in $U \setminus S$). Noting that $\varphi \circ f - \varphi \circ f \circ h$ is an element of $N_0^{1,Q}(U)$, we see that $\varphi \circ f = \varphi \circ f \circ h$. Hence, in this case, $\varphi \circ f(y)$ depends only on $\Pi_i(y)$ for $y \in V_i$ and we can show as before that the corresponding function on e_i is linear.

By applying Proposition 5.8 to the discussion in the previous two paragraphs, we obtain:

Proposition 5.18. *Each $x \in U$ is contained in a neighborhood contained in U that is a union $\bigcup_{r=1}^{N} V_i^r$ with $V_i^r = \Pi_i^{-1}(e_i^r)$ and $e_i^r \subseteq X_i$ distinct edges ($N > 1$ if and only if $x \notin S$ is a vertex level, and the e_i^r are the edges adjacent to $\Pi_i(x)$), such that the restriction of f to each $\tilde{V}_i^r = \Pi_i^{-1}(\text{interior}(e_i^r))$ is an open embedding that preserves differences in height up to the constant $\lambda_r \in \{\pm m^{-\ell_r}\}$ with $\ell_r \in \mathbb{Z}$.*

We can say more about ℓ_r if $N > 1$:

Proposition 5.19. *If $N > 1$ in Proposition 5.18, then ℓ_r does not depend on r.*

Proof. Assume $m \geq 3$. Recall that $V_i^r = \Pi_i^{-1}(e_i^r)$ and let $V = \bigcup_{r=1}^{N} V_i^r$, so that V is a neighborhood of x and any other y with $\Pi_i(x) = \Pi_i(y)$.

Let $\ell = \min_r \ell_r$. For each e_i^r such that $\ell_r > \ell$, since $f(V_i^r)$ is the preimage under $\Pi_{i+\ell_r}$ of an edge of $X_{i+\ell_r}$, we see that $\Pi_{i+\ell+1}(f(V_i^r))$ is contained in an edge of $X_{i+\ell+1}$.

Consider the union W of the $2 \cdot k \cdot k$ edges in $X_{i+\ell+1}$ that intersect $\pi_{i+\ell}^{-1}(\Pi_{i+\ell}(f(x)))$. By Corollaries 5.9 and 5.10, for each e_i^r with $\ell_r = \ell$, we have that $\Pi_{i+\ell+1}(f(V_i^r))$ intersects the interior of k edges in W, and each of these k edges intersects a unique point in $\pi_{i+\ell}^{-1}(\Pi_{i+\ell}(f(x)))$. For each e_i^r with $\ell_r > \ell$, we have that $\Pi_{i+\ell+1}(f(V_i^r))$ intersects the interior of only one edge in W, since $f(V_i^r)$ is the preimage under $\Pi_{i+\ell_r}$ of an edge of $X_{i+\ell_r}$, and so we see that $\Pi_{i+\ell+1}(f(V_i^r))$ is contained in an edge of $X_{i+\ell+1}$. Thus, if there exists $\ell_r > \ell$, then $\Pi_{i+\ell+1} \circ f$ cannot be an open map: there is a neighborhood of x in V which is mapped to a set under $\Pi_{i+\ell+1} \circ f$ that contains $\pi_{i+\ell}^{-1}(\Pi_{i+\ell}(f(x)))$ but cannot contain a neighborhood of some point of $\pi_{i+\ell}^{-1}(\Pi_{i+\ell}(f(x)))$, as $\Pi_{i+\ell+1}(f(V))$ can only intersect the interiors of $< 2 \cdot k \cdot k$ edges in W. This is a contradiction noting that $\Pi_{i+\ell+1}$ and f are open maps.

If $m = 2$, we either apply the same argument as above or, if necessary, replace the e_i^r with the edges e_{i+1}^r which make up the star of $\Pi_{i+1}(x)$ in X_{i+1}. □

The following theorems are easy consequences of Propositions 5.18 and 5.19:

Theorem 5.20. *Let $f : U \to V$ be a quasiconformal homeomorphism between connected open subsets of X_∞. Then there exists $\ell \in \mathbb{Z}$ such that f is locally an $m^{-\ell}$-homothety. In particular, for each $x \in U$, there exists i and $E \subseteq U$ with $E = \bigcup_{r=1}^{N} \Pi_i^{-1}(e_i^r)$, where e_i^r are the edges in X_i intersecting $\Pi_i(x)$, such that the restriction of f to each $\Pi_i^{-1}(e_i^r)$ is an $m^{-\ell}$-homothety between $\Pi_i^{-1}(e_i^r)$ and an edge preimage of level $i + \ell$.*

Theorem 5.21. *Let $f : X_\infty \to X_\infty$ be a quasiconformal homeomorphism. Then f is an isometry. Moreover, there exists i such that the restriction of f to any edge preimage of level i is an isometry onto an edge preimage of level i.*

Theorem 5.22. *Let $f : X_\infty \setminus S \to X_\infty$ be an open quasiconformal embedding.*

(1) If $m \geq 3$, then f is an m^{-i}-homothety onto some edge preimage of level i.

(2) *If* $m = 2$ *and the conclusion of (1) does not hold, then* f *is still an* m^{-i}*-homothety in this case. The restriction of* f *to any edge preimage of level* 1 *is an* m^{-i}*-homothety onto some edge preimage of level* $1 + i$*.*

In each of the previous three theorems, we may apply Corollaries 5.9 and 5.10 to obtain a description of f as an inverse limit of isomorphisms between graphs.

Note, in a preprint from 2001 ([17]), T. Laakso states that he can use tangent cones to prove that bi-Lipschitz and quasiconformal maps between open subsets of Laakso spaces are local homotheties, though to the author of this dissertation's knowledge such a result has not appeared in print.

6 Right-angled Fuchsian buildings

In this section, we discuss applications of the above ideas to the boundaries of Fuchsian buildings; in particular we prove that quasiconformal homeomorphisms between open subsets preserve Q-energy and Q-harmonic functions. We also give conditions which imply that Q-harmonic functions descend to graphs, specifically tree-walls.

6.1 Preliminaries

We refer the reader to [2], [3], [4], or [25] for background on Fuchsian buildings, though only the last reference defines them in the most general form. For our purposes, we define a *right-angled Fuchsian building with constant thickness* q as follows: Fix integers $p \geq 5$ and $q \geq 3$. Let R denote the unique (up to isometry) right-angled regular p-gon in the hyperbolic plane (with constant curvature -1). The building $I_{p,q}$ is defined to be the unique simply-connected 2-complex with each 2-cell isomorphic to R and with the link of each vertex isomorphic to the complete (q, q)-bipartite graph (uniqueness is proven in [2]).

We call the 2-cells in $I_{p,q}$ *chambers*. Let A denote the 2-complex obtained by tiling the hyperbolic plane with isometric copies of R. An *apartment* of $I_{p,q}$ is a subcomplex isomorphic to A. Then:

(1) Any two chambers in $I_{p,q}$ are contained in a common apartment.

(2) Given any two apartments A_1, A_2 in $I_{p,q}$ that share a common chamber, there is a cellular isomorphism $A_1 \to A_2$ which fixes $A_1 \cap A_2$ pointwise.

We note that the length metric on $I_{p,q}$ makes it a CAT(-1) space, hence a Gromov hyperbolic metric space, with Gromov boundary at infinity homeomorphic to the Menger curve.

We assume that integers $p \geq 5$ and $q \geq 3$ are fixed for the remainder of this section.

Given a chamber c and an apartment A containing c, the retraction onto A centered at c is a cellular map $r = r_{c,A} : I_{p,q} \to A$ defined as follows: Suppose that c' is a chamber in $I_{p,q}$. Then there is an apartment A' containing both c and c', and there is a cellular isomorphism $f : A' \to A$ which fixes $A' \cap A$ pointwise. Let $r \mid c' = f \mid c'$, and it can be shown that r is well-defined.

Fixing a chamber c, we refer the reader to [4], Section 2, for the definition of the length metric d_c on the Gromov boundary $\partial I_{p,q}$ of $I_{p,q}$ (the notation δ_c is used in this reference).

In [3], it is shown that $(\partial I_{p,q}, d_c, \mathcal{H}^Q)$ is a PI space; this result can be extended to general Fuchsian buildings. In fact, $(\partial I_{p,q}, d_c, \mathcal{H}^Q)$ is compact, geodesic, Ahlfors Q-regular for some $Q > 1$, and admits a $(1,1)$-Poincaré inequality. Let $r = r_{c,A}$ as above and $\rho = \partial r : \partial I_{p,q} \to \partial A$ denote the boundary map of r. Then ρ is a retraction, and observing that ∂A is isometric to a circle, it is known that ρ is a "global 1-dimensional differentiability chart" for $\partial I_{p,q}$ (this fact does not appear in the literature to the author's knowledge). We put quotes because, technically, we need to modify the domain and codomain of ρ so that its image is an interval (which covers $\partial I_{p,q}$ with two charts), though we just abuse notation and say that ρ is a differentiability chart.

We remark that in [4], it is shown that any quasi-isometry of a right-angled Fuchsian building lies within bounded distance of an isometry, and in [25], this result is extended to all Fuchsian buildings.

6.2 Preservation of Q-energy and Q-harmonic functions

Given a base chamber $c \subseteq I_{p,q}$ and an apartment A containing c, let $Q > 1$ denote the Hausdorff dimension of $(\partial I_{p,q}, d_c)$. Let $r = r_{c,A} : I_{p,q} \to A$ denote the retraction onto A centered at c and $\rho = \partial r : \partial I_{p,q} \to \partial A$ the boundary map of r.

Suppose that $f : U \to V$ is a quasiconformal homeomorphism between open subsets of $\partial I_{p,q}$. In [4], a differential of f for a.e. $x \in U$ is defined as follows: Choose an apartment A' containing c such that $x \in \partial A'$. Parameterizing $\partial A'$ (which is a circle) by arc-length (so $x \pm t$ is the point a

distance t from x in $\partial A'$ in the "±"-direction), we define

$$|f'(x)| = \lim_{t \to 0} d_c(f(x+t), f(x))/|t|.$$

Using the fact that ρ is a differentiability chart, it is easy to see that $|f'(x)| = |Df(x)|$ for a.e. $x \in \partial I_{p,q}$, where $|Df(x)|$ is defined as in Section 3.2.

By [4], Proposition 3.2.1, we have

$$|Df(x)| = \mathrm{Lip}_x f$$

for a.e. $x \in U$ (this proposition is stated for global quasiconformal homeomorphisms, but the proof still works for the local case). We can use this fact and adapt the proof of Theorems 4.2 and 4.3 (preservation of Newtonian Q-energy and preservation of Q-harmonic functions) to obtain the following:

Theorem 6.1. *Let $f : U \to V$ be a quasiconformal homeomorphism between open subsets of $\partial I_{p,q}$. If $u \in N^{1,Q}(V)$ is bounded, then $u \circ f \in N^{1,Q}(U)$ and*

$$\|\rho_{u \circ f}\|_{L^Q(U)} = \|\rho_u\|_{L^Q(V)}.$$

If, in addition, u is bounded and Q-harmonic, then $u \circ f$ is Q-harmonic.

6.3 Q-harmonic functions

Our first objective is to prove an analogue of Proposition 5.6 for $\partial I_{p,q}$. We need some definitions first.

A *wall* in $I_{p,q}$ is a geodesic line contained in the 1-skeleton. We obtain an equivalence relation by saying that two edges of $I_{p,q}$ are equivalent if they are contained in the same wall. An

equivalence class of this equivalence relation is called a *tree-wall*.

The following subcomplexes of $I_{p,q}$ are defined in [2], Section 2.2: Let $I(1) = c$. Inductively define $I(n+1)$ to be the union the cells in $I_{p,q}$ that have at least one vertex in common with $I(n)$.

Given a vertex s of $\partial I(n)$, we say that s is *free* if it is only incident with one cell of $I(n)$; otherwise it is *partially free* and it is incident with q cells of $I(n)$ that are joined along an edge, and s is one of the vertices of this edge. Then $I(n+1)$ is obtained from $I(n)$ in a two-step procedure:

(1) To every edge of $\partial I(n)$, glue $(q-1)$ p-gons (along an edge).

(2) At every partially free vertex of $\partial I(n)$, group the incident p-gons attached in (1) into groups of q p-gons, ensuring no two of the p-gons in a group are currently attached to the same edge. For the polygons in each group, glue them along their edges incident with the partially free vertex and not contained in $\partial I(n)$.

(2′) At every free vertex, group the incident p-gons attached in (1) into groups of 2 p-gons, ensuring the two p-gons in a group are currently attached to different edges. For each group, attach a new p-gon so that the link of the subcomplex consisting of the 2-cell in $I(n)$ incident with the free vertex, the two p-gons in the group, and the new p-gon is a circle.

(Technically speaking, we have described how a complex isomorphic to $I(n+1)$ is obtained from a complex isomorphic to $I(n)$). Note that in an edge of $\partial I(n)$, at least one of its vertices must be free.

Using the construction above, we may describe how to extend an isomorphism $f_n : I(n) \to I(n)$ to an isomorphism $f_{n+1} : I(n+1) \to I(n+1)$:

(C1) At each partially free vertex of $\partial I(n)$, there are $q-1$ edges of $I_{p,q}$ not contained in $I(n)$. If x is a partially free vertex of $\partial I(n)$, then so is $f_n(x)$, and we let E and E' respectively denote the sets of edges corresponding to x and $f_n(x)$ as described in the previous sentences. We

may assign any bijection $E \to E'$ to f_{n+1}, which determines f_{n+1} on the cells that contain these edges.

(C2) At each edge whose two vertices are free, there are $q-1$ chambers of $I_{p,q}$, not contained in $I(n)$, that contain that edge. If e is such an edge, then so is $f_n(e)$, and we let the sets F and F' respectively denote the corresponding 2-cells as described in the previous sentence. We may assign any bijection $F \to F'$ to f_{n+1}.

(C3) f_{n+1} on the remaining cells of $I(n+1)$ is determined by the previous choices.

We describe how to construct an isometry of $I_{p,q}$ that swaps two chambers incident with a single edge.

Proposition 6.2. *Let a, a_1, and a_2 be distinct chambers of $I_{p,q}$ incident with a single edge e. Let T be the tree-wall containing e. Then there is an isometry of $I_{p,q}$ mapping a_1 onto a_2 and fixing pointwise the component of T^c containing a.*

Let $I(n)$ denote the increasing sequence of subcomplexes described at the beginning of this subsection, using a as the base chamber. The proposition follows from the following lemma, after observing that there is an isometry $I(2) \to I(2)$ mapping a_1 onto a_2 and fixing pointwise the component of $T^c \cap I(2)$ containing a.

Lemma 6.3. *Suppose that we are given an isometry $f_n : I(n) \to I(n)$ mapping a_1 onto a_2 and fixing pointwise the component of $T^c \cap I(n)$ containing a, where $n \geq 2$. There exists an extension $f_{n+1} : I(n+1) \to I(n+1)$ fixing pointwise the component of $T^c \cap I(n+1)$ containing a.*

Proof. An inductive argument shows that $T \cap \partial I(n)$ consists of partially-free vertices only. In fact, $T \cap \partial I(2)$ consists of partially-free vertices only; if $n \geq 3$, for every partially-free vertex of $\partial I(n-1)$, for any edge incident with that vertex and $\partial I(n)$, its vertex incident with $\partial I(n)$ is a partially-free vertex of $\partial I(n)$.

Let E be the edges of $I(n+1)$ incident with a partially-free vertex of $\partial I(n)$ but not contained in $I(n)$. By (C1), we may prescribe f_{n+1} to be the identity on those edges of E incident with T or contained in the same component of $T^c \cap I(n+1)$ as a. Similarly, at each edge of $\partial I(n)$ whose two vertices are free, we may prescribe f_{n+1} to be the identity on all chambers containing that edge by (C2). These choices together determine f_{n+1} to be the identity on all of the chambers of $T^c \cap I(n+1)$ in the same component as a. We finish on the remaining chambers of $I(n+1)$ by making any choices as in (C1), (C2). □

After the following lemma, we prove an analogue of Proposition 5.6 for $\partial I_{p,q}$.

Lemma 6.4. *Let T be a tree-wall in $I_{p,q}$. Let $\gamma : [0,\infty) \to I_{p,q}$ be a geodesic ray with $\gamma(0) \in T$ so that $\gamma(0)$ is not a vertex and that γ leaves T orthogonally. If $\gamma(t)$ lies on a tree-wall T' for some $t > 0$, then T' does not intersect T.*

Proof. If T' intersects T, we may form a triangle in $I_{p,q}$ formed with three geodesic segments joining a, b, c pairwise, with $a = \gamma(0)$, $b = \gamma(t)$, and c a point in $T' \cap T$. The segment from b to c lies inside T' and the segment from c to a lies in T, since tree-walls are convex; the segment from a to b is along the geodesic ray γ. Noting that distinct tree-walls intersect orthogonally, we see that the sum of the interior angles of the triangle we have formed is greater than 2π, which cannot occur in a CAT(0) space. □

Proposition 6.5. *Fix a base chamber c in $I_{p,q}$. Let T be a tree-wall and suppose that γ_1, γ_2 are two geodesic rays not intersecting the component of T^c containing c. Further suppose that $\gamma_1(0) = \gamma_2(0)$, that $\gamma_1(0)$ is not a vertex, and that γ_1, γ_2 leave T orthogonally. Then there is an isometry of $I_{p,q}$ mapping γ_1 onto γ_2 that fixes pointwise the component of T^c containing c.*

Proof. Let $0 < t_1 < t_2 < \cdots$ be the sequence of times such that $\gamma_2(t_i)$ intersects a tree-wall. We prove, by induction on n, that there is an isometry f_n such that $f_n \circ \gamma_1$ agrees with γ_2 on $[0, t_n]$ and such that f_n fixes pointwise the component of T^c containing c. The proposition follows by taking a subsequential limit of the f_n.

The base case $n = 1$ is handled by Lemma 6.3.

Assume we have such an f_n for $n \geq 1$. There are two cases: (1) $\gamma_2(t_n)$ is not a vertex and (2) $\gamma_2(t_n)$ is a vertex.

(1) Consider the tree-wall T_n containing $\gamma_2(t_n)$. By Lemma 6.4, T_n does not intersect T, which implies that the component of T_n^c containing c contains the component of T^c containing c. We may use Lemma 6.3 to find an isometry f'_{n+1}, which is the identity on the component of T_n^c containing c, so that $f'_{n+1} \circ f_n \circ \gamma_1$ agrees with γ_2 up to t_n. We let $f_{n+1} = f'_{n+1} \circ f_n$.

(2) There are seven chambers under consideration: $w, x_1, x_2, y_1, y_2, z_1, z_2$. We define w to be the chamber in which $\gamma_2(t)$ lies for close $t < t_n$; x_1, x_2 to be the chambers in which $f_n \circ \gamma_1(t), \gamma_2(t)$ lie respectively for close $t > t_n$; and y_1, z_1 (resp. y_2, z_2) chosen to connect w to x_1 (resp. x_2) in the link of $\gamma_2(t_n)$, so that y_1, y_2 (resp. z_1, z_2) share an edge e_1 (resp. e_2) contained in w. Let T_n^1 (resp. T_n^2) be the tree-wall containing e_1 (resp. e_2).

By similar reasoning as in (1), the component of $(T_n^1)^c$ containing c also contains the component of T^c containing c; this former component also contains z_1, z_2, and $\gamma_2([0, t_n])$. We can then find an isometry f_{n+1}^1 fixing pointwise $z_1, z_2, \gamma_2([0, t_n])$, and the component of T^c containing c, but mapping y_1 onto y_2. Similarly, there is an isometry f_{n+1}^2 fixing pointwise $y_1, y_2, \gamma_2([0, t_n])$, and the component of T^c containing c, but mapping z_1 onto z_2. Then $f_{n+1}^1 \circ f_{n+1}^2$ maps x_1 onto x_2 and fixes pointwise the component of T^c containing c. We let $f_{n+1} = f_{n+1}^1 \circ f_{n+1}^2 \circ f_n$. □

We can now state how Q-harmonic functions descend to graphs.

Theorem 6.6. *Let $U \subseteq \partial I_{p,q}$ be open and $u : U \to \mathbb{R}$ a Q-harmonic function. Suppose that T is a tree-wall with $\partial T \subseteq U$. Let U' be the boundary of the component of T^c containing c. Suppose that U' contains U^c. Let π be the boundary map of the nearest-point projection to T. Then, for $x \in (U')^c$, we have that $u(x)$ only depends on $\pi(x)$.*

Proof. Since the boundaries of tree-walls are $\mathcal{H}^Q$-null, for a dense set of x we have that $\pi(x)$ is in the interior of an edge of T. Let $x, y \in (U')^c$ be such that $\pi(x) = \pi(y)$ is not a vertex of T.

Consider the geodesic-rays $\gamma_1, \gamma_2 : [0, \infty) \to I_{p,q}$ beginning at $\pi(x)$ with boundary values x, y respectively.

By Proposition 6.5, there is an isometry f of $I_{p,q}$ mapping γ_1 onto γ_2 and such that ∂f is an isometry of $\partial I_{p,q}$ mapping x to y that is the identity on U'. The theorem follows by the uniqueness part of Theorem 2.7. □

References

[1] Björn, A. and Björn, J. (2011). *Nonlinear Potential Theory on Metric Spaces*, European Mathematical Society (EMS) Press. https://doi.org/10.4171/099

[2] Bourdon, M. (1997). Immeubles hyperboliques, dimension conforme et rigidité de Mostow. *Geom. funct. anal.* 7, 245–268. https://doi.org/10.1007/PL00001619

[3] Bourdon, M. and Pajot, H. (1999). Poincaré Inequalities and Quasiconformal Structure on the Boundary of Some Hyperbolic Buildings. *Proc. Amer. Math. Soc.* 127, 2315–2324. http://www.jstor.org/stable/119267

[4] Bourdon, M. and Pajot, H. (2000). Rigidity of quasi-isometries for some hyperbolic buildings. *Comment. Math. Helv.* 75, 701–736. https://doi.org/10.1007/s000140050146

[5] Capogna, L., and Cowling, M. (2006). Conformality and Q-harmonicity in Carnot groups. *Duke Math. J.* 135, 455–479. https://doi.org/10.1215/S0012-7094-06-13532-9

[6] Cowling, M. G. and Ottazzi, A. (2015). Conformal maps of Carnot groups. *Ann. Acad. Sci. Fenn. Math.* 40, 203–213. https://doi.org/10.5186/aasfm.2015.4008

[7] Cheeger, J. (1999). Differentiability of Lipschitz Functions on Metric Measure Spaces. *Geom. funct. anal.* 9, 428–517. https://doi.org/10.1007/s000390050094

[8] Cheeger, J. and Kleiner, B. (2015). Inverse limit spaces satisfying a Poincaré inequality. *Analysis and Geometry in Metric Spaces*, 3 (1), 15–39. https://doi.org/10.1515/agms-2015-0002

[9] Cheeger, J., Kleiner, B., and Schioppa, A. (2016). Infinitesimal structure of differentiability spaces, and metric differentiation. *Analysis and Geometry in Metric Spaces*, 4 (1), 104–159. https://doi.org/10.1515/agms-2016-0005

[10] Heinonen, J. (2001). *Lectures on analysis on metric spaces.* Springer. https://doi.org/10.1007/978-1-4613-0131-8

[11] Heinonen, J. and Koskela, P. (1998). Quasiconformal maps in metric spaces with controlled geometry. *Acta Math.* 181, 1–61. https://doi.org/10.1007/BF02392747

[12] Keith, S. and Zhong, X. (2008). The Poincaré inequality is an open ended condition. *Ann. of Math.* 167 (2), 575–599. https://doi.org/10.4007/annals.2008.167.575

[13] Kinnunen, J. and Shanmugalingam, N. (2001). Regularity of quasi-minimizers on metric spaces. *Manuscripta math.* 105, 401–423. https://doi.org/10.1007/s002290100193

[14] Kleiner, B. (2007). The asymptotic geometry of negatively curved spaces: uniformization, geometrization and rigidity. *Proceedings of the International Congress of Mathematicians Madrid, August 22–30, 2006*, 743–768. https://doi.org/10.4171/022-2/36

[15] Kleiner, B. and MacKay, J. M. (2016). Differentiable structures on metric measure spaces: a primer. *Ann. Sc. Norm. Super. Pisa Cl. Sci.* 16 (1), 41–64. https://doi.org/10.2422/2036-2145.201403_004

[16] Laakso, T. (2000). Ahlfors Q-regular spaces with arbitrary Q > 1 admitting weak Poincaré inequality. *Geom. funct. anal*. 10, 111–123. https://doi.org/10.1007/s000390050003

[17] Laakso, T. (2001). Tangent Cones of Certain Spaces Admitting Poincaré Inequality. Preprint 292. Department of Mathematics, University of Helsinki. (2001). https://wiki.helsinki.fi/xwiki/bin/view/mathstat/Tutkimus/Reports/

[18] Laakso, T. (2002). Erratum to "Ahlfors Q-regular spaces with arbitrary Q > 1 admitting weak Poincaré inequality", in GAFA 10 (2000). *Geom. funct. anal*. 12, 650. https://doi.org/10.1007/s00039-002-8261-9

[19] Mostow, G. D. (1968). Quasi-conformal mappings in n-space and the rigidity of hyperbolic space forms. *Publ. Math. Inst. Hautes Études Sci.* 34, 53–104. https://doi.org/10.1007/BF02684590

[20] Pansu, P. (1989). Métriques de Carnot-Carathéodory et quasiisométries des espaces symétriques de rang un. *Ann. of Math.* 129, 1–60. https://doi.org/10.2307/1971484

[21] Shanmugalingam, N. (2000). Newtonian spaces: An extension of Sobolev spaces to metric measure spaces. *Rev. Mat. Iberoam.* 16 (2), 243–279. https://doi.org/10.4171/rmi/275

[22] Shanmugalingam, N. (2001). Harmonic functions on metric spaces. *Illinois J. Math.* 45 (3), 1021–1050. https://doi.org/10.1215/ijm/1258138166

[23] Tyson, J. T. (2001). Metric and geometric quasiconformality in Ahlfors regular Loewner spaces. *Conform. Geom. Dyn.* 5, 21–73. https://doi.org/10.1090/S1088-4173-01-00064-9

[24] Väisälä, J. (1981). Quasisymmetric embeddings in Euclidean spaces. *Trans. Amer. Math. Soc.* 264, 191–204. https://doi.org/10.1090/S0002-9947-1981-0597876-7

[25] Xie, X. (2006). Quasi-isometric rigidity of Fuchsian buildings. *Topology.* 45, 101–169. https://doi.org/10.1016/j.top.2005.06.005